Radxa ROCK 5B

schematic

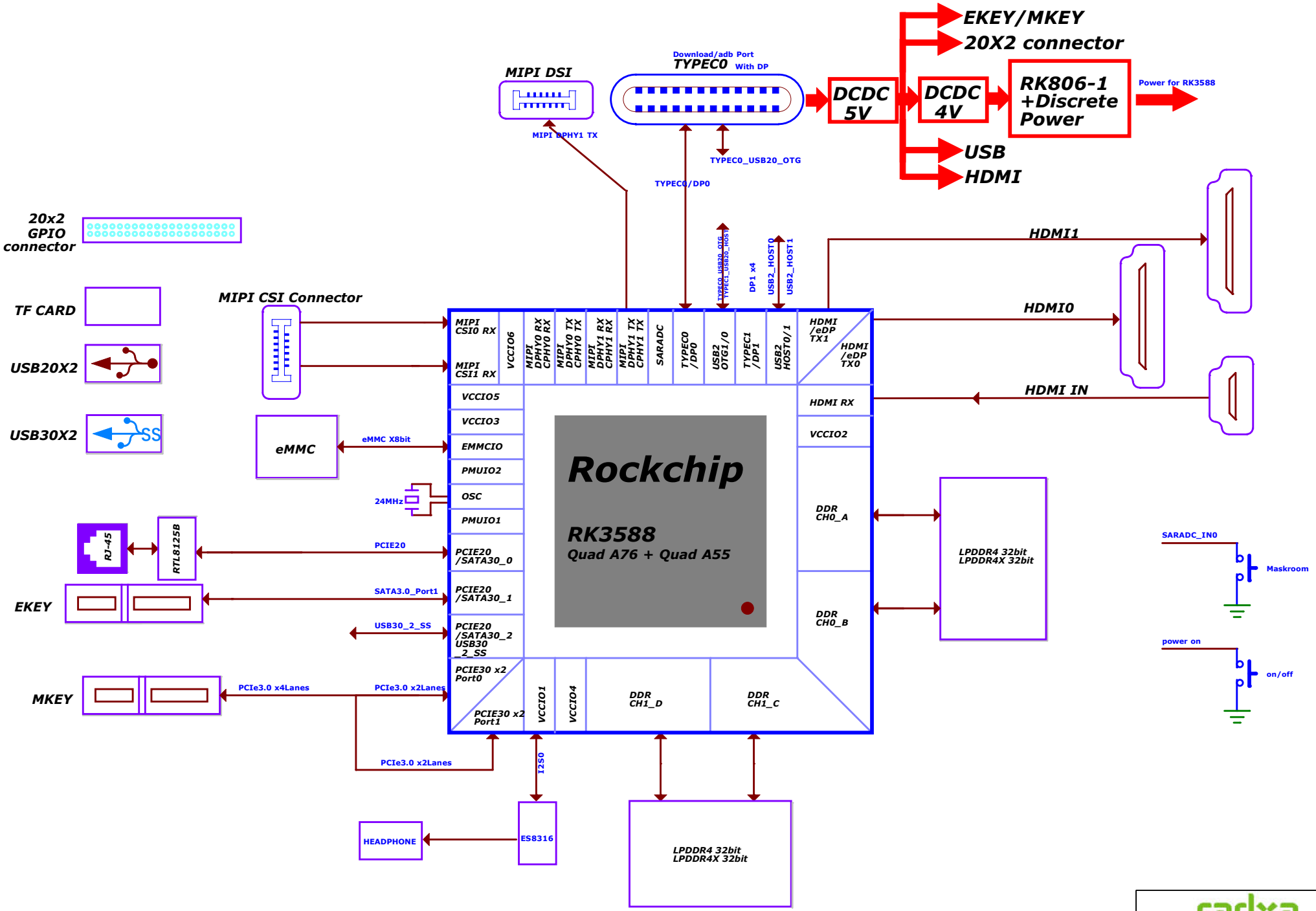
V1.45

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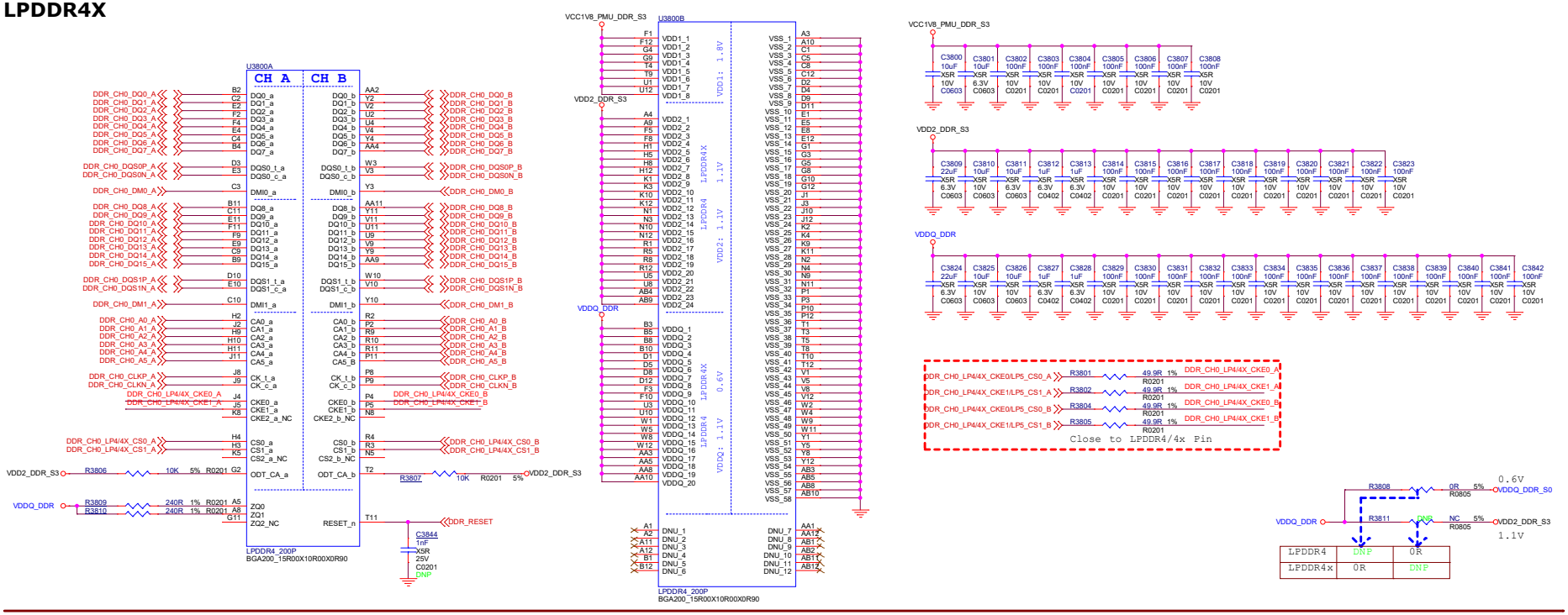
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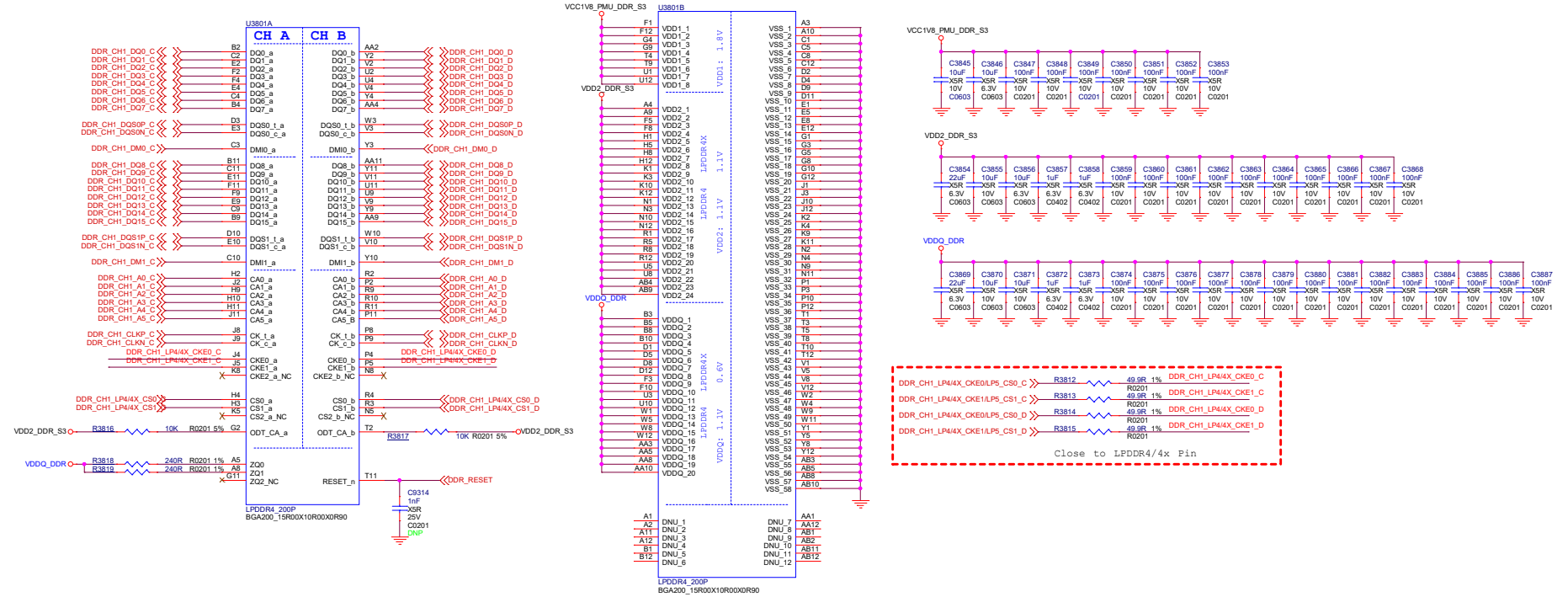


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A2	Page Name:	Block Diagram	1.45
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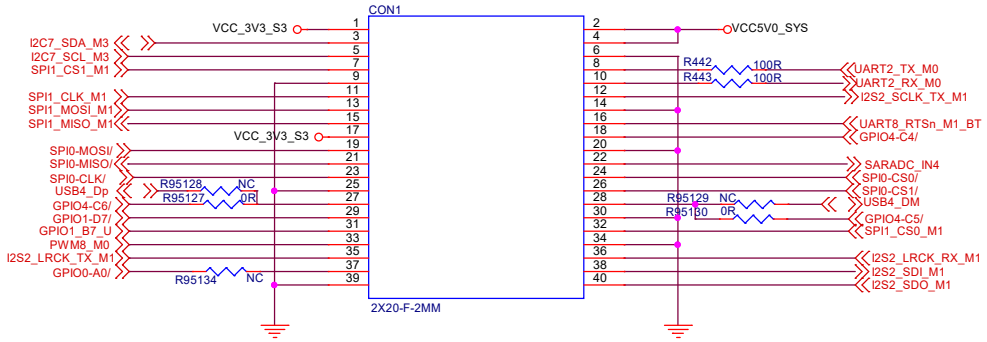
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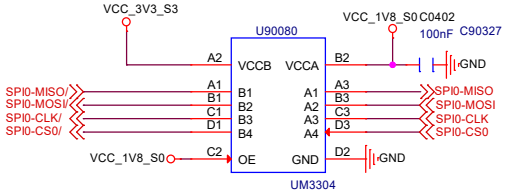


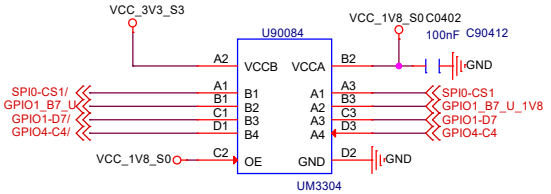


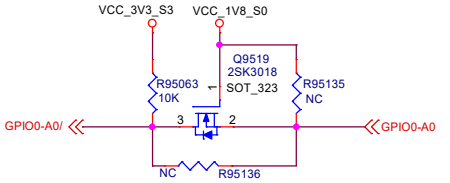
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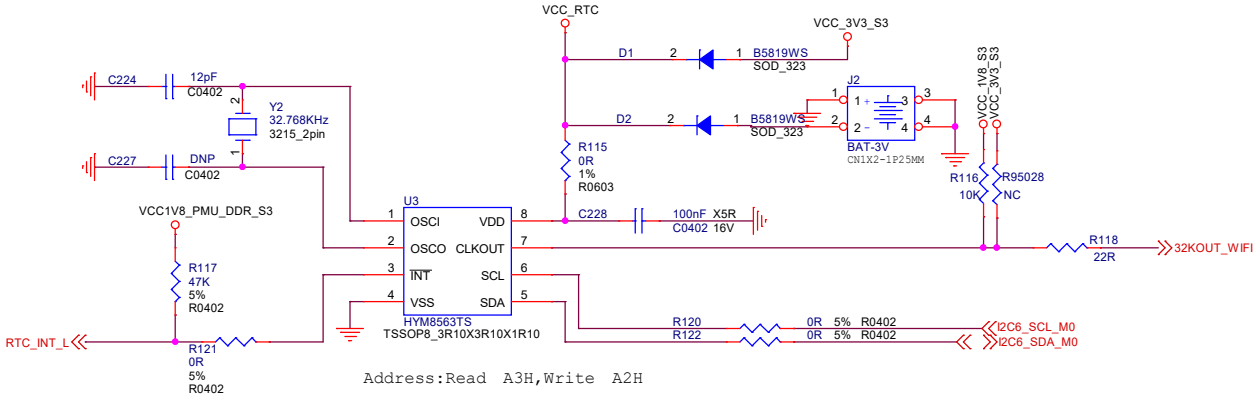
CONNECT

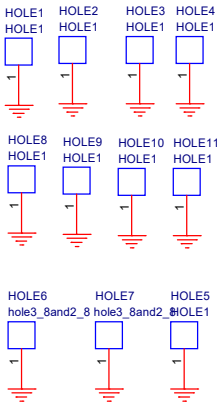






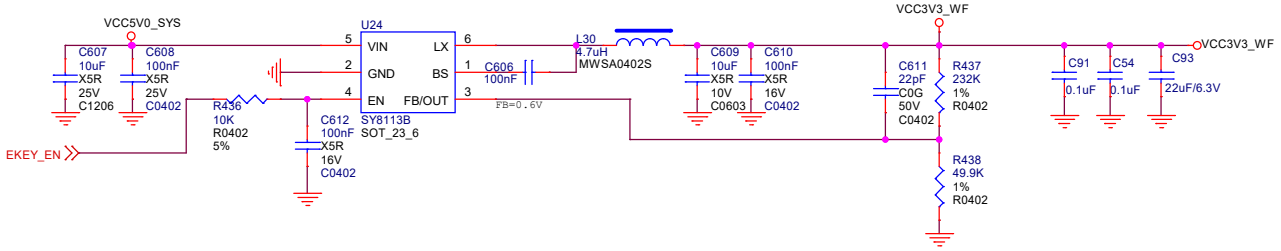


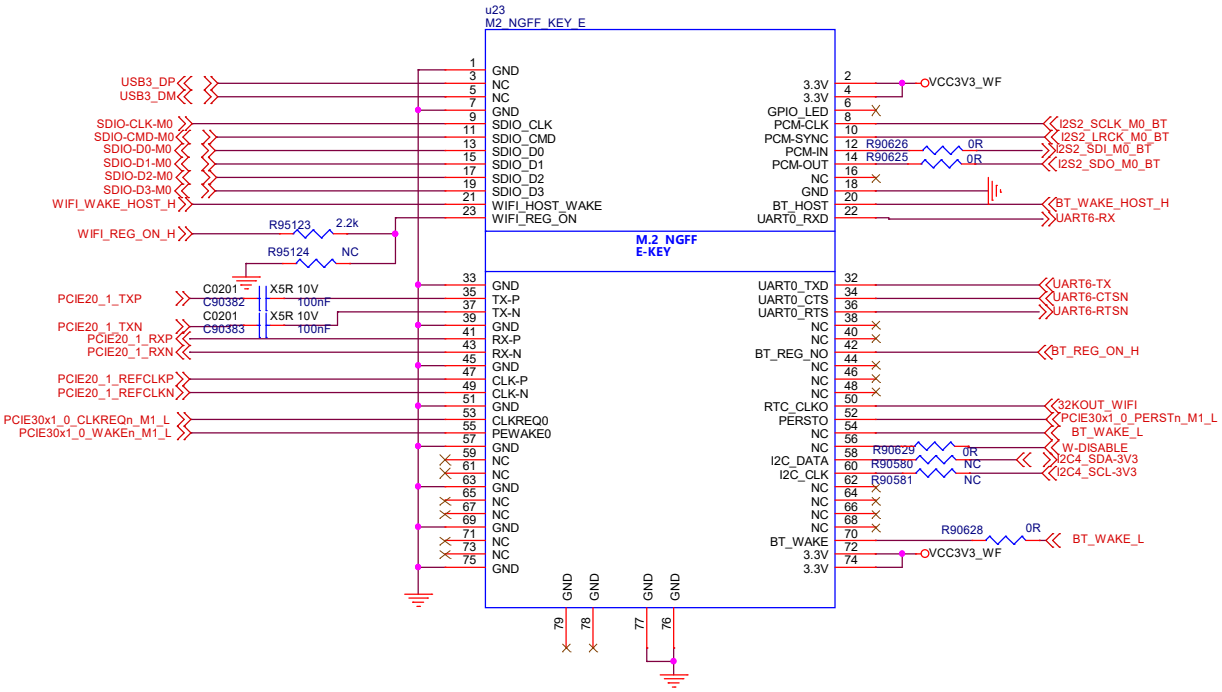




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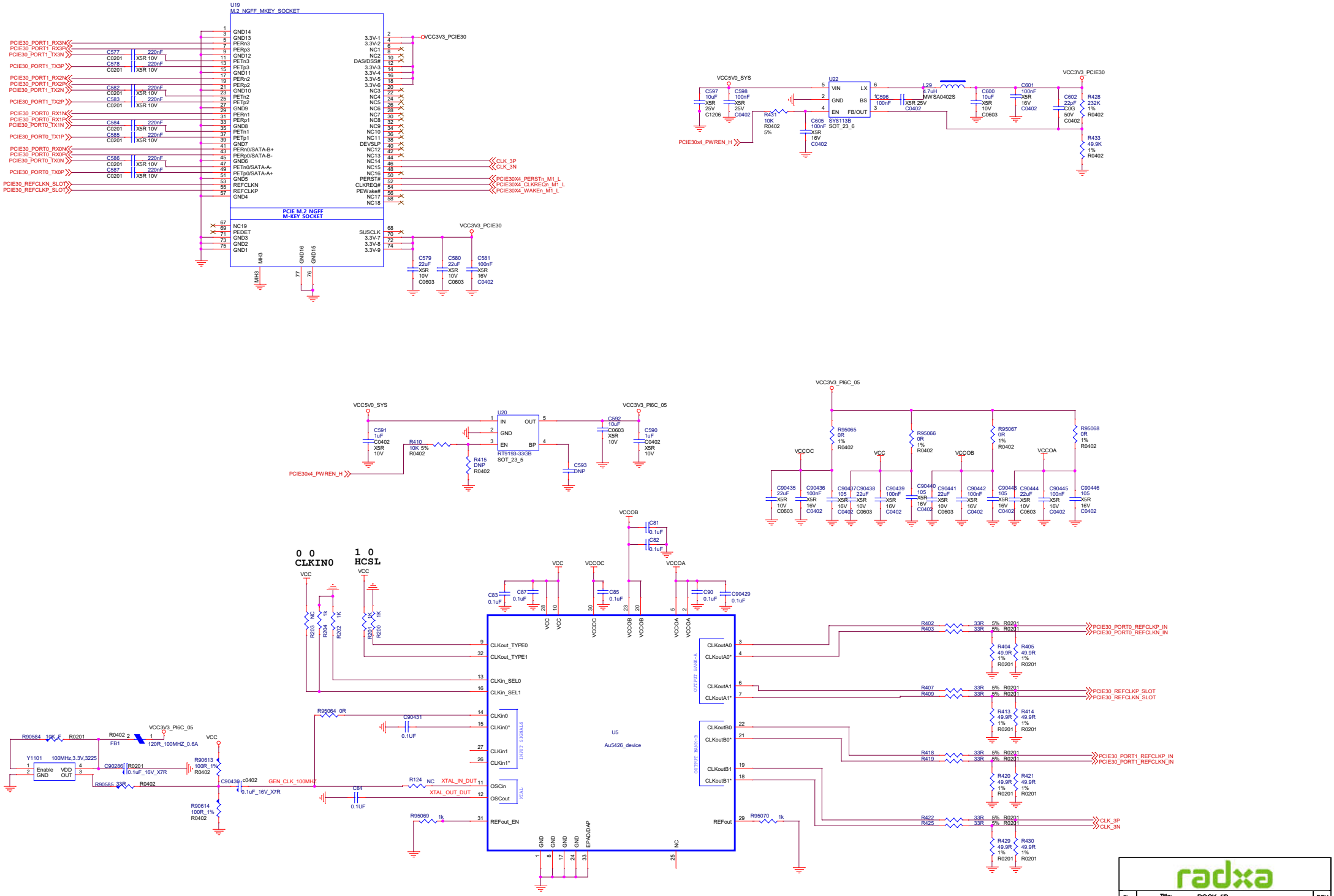
PCIE-PCIE2.0_Slot-Ekey





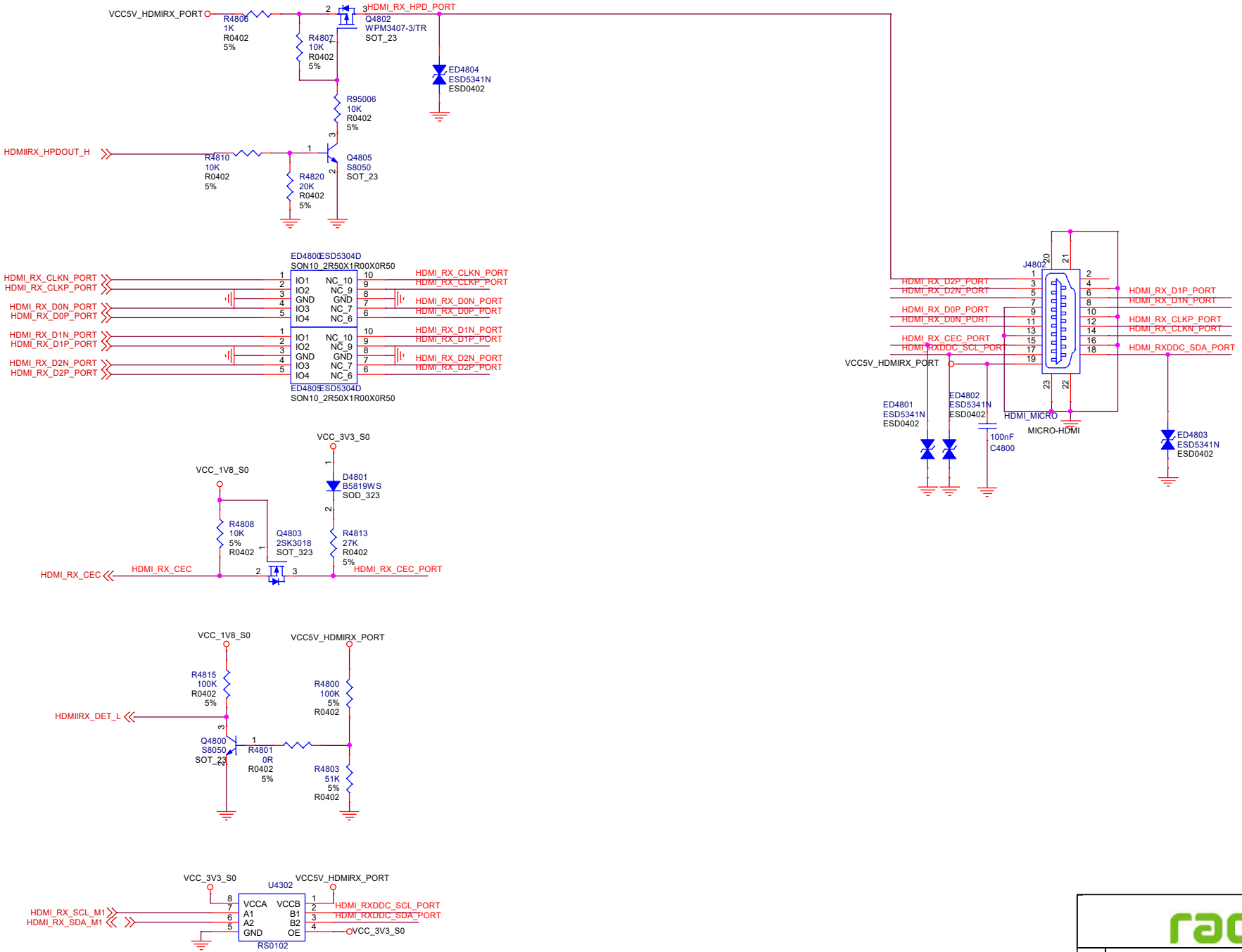
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A3	Page Name: PCIE-PCIE2.0_Slot-Ekey		1.45
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PCIe3.0 x 4 Slot



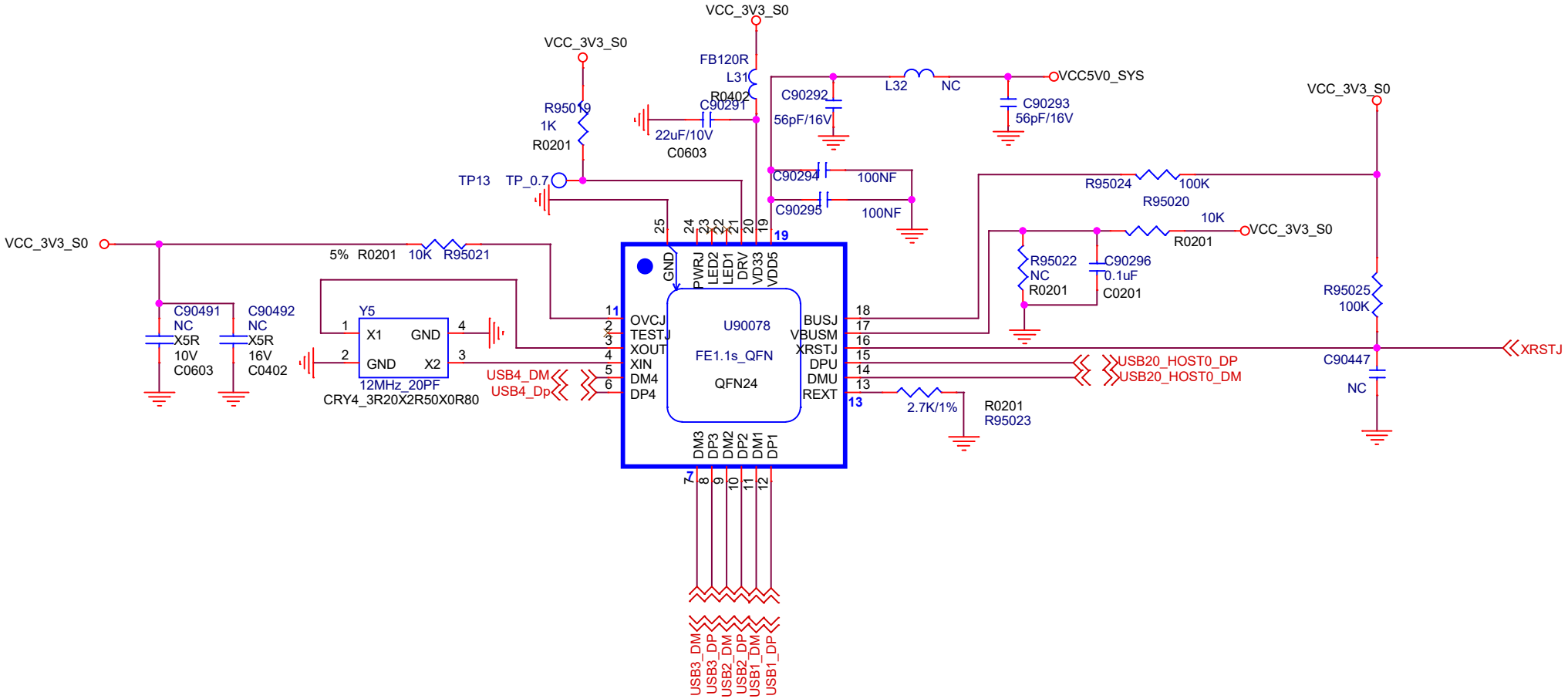
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HDMI2.0 RX

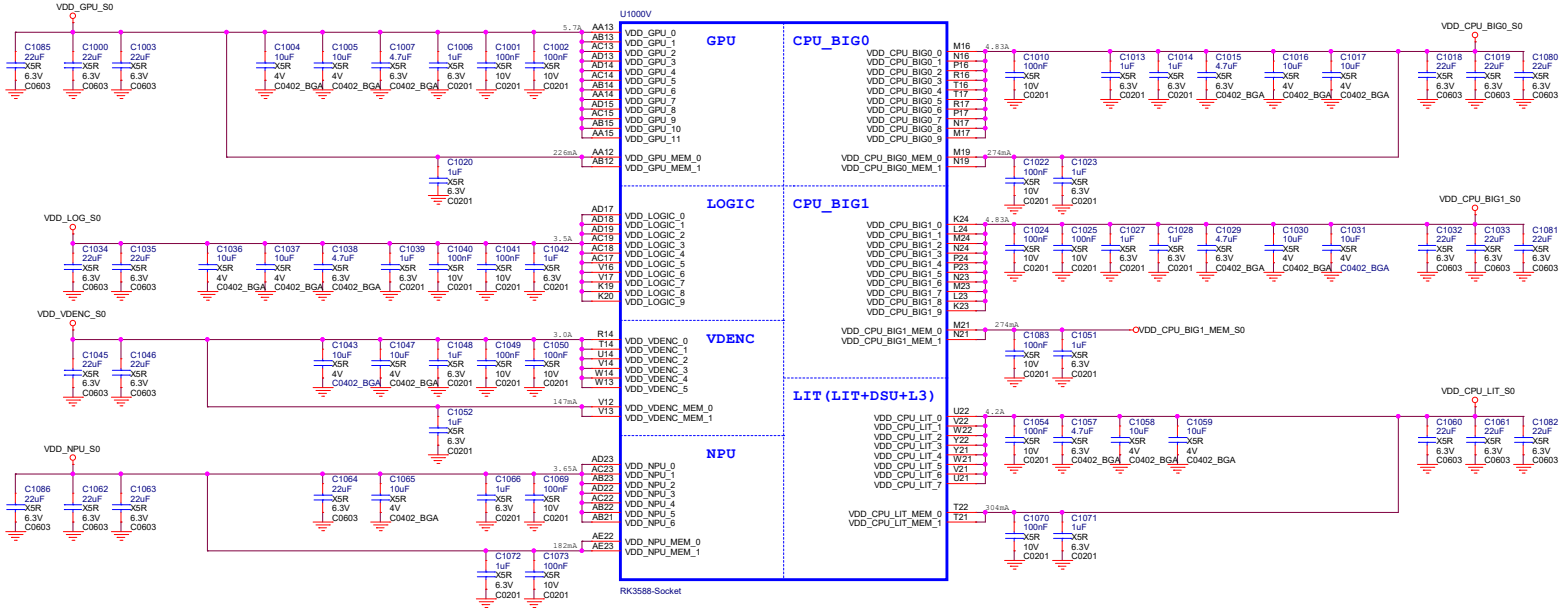


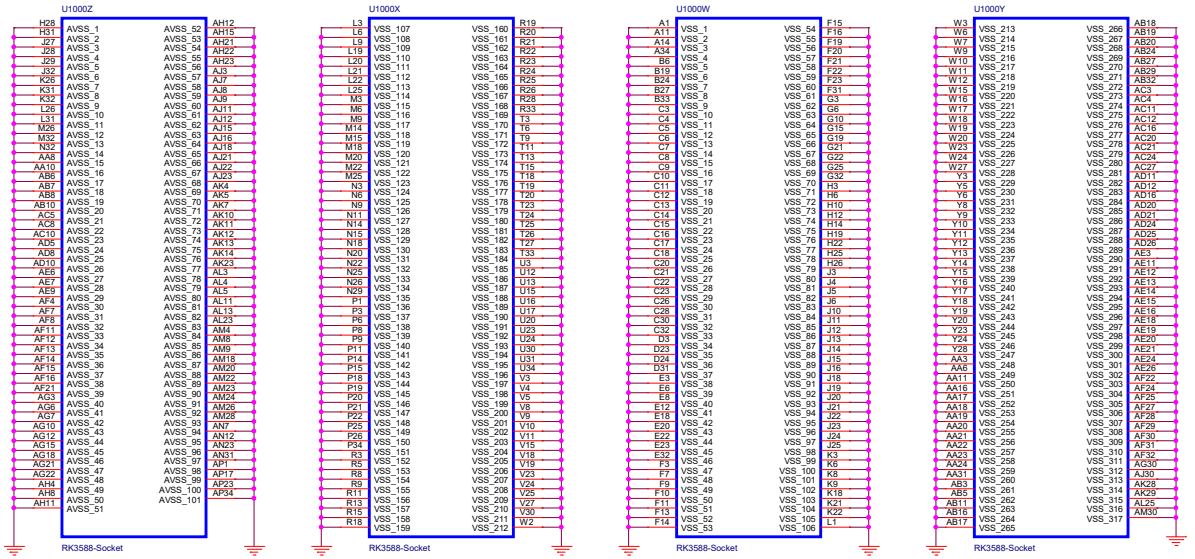
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USB2HUB



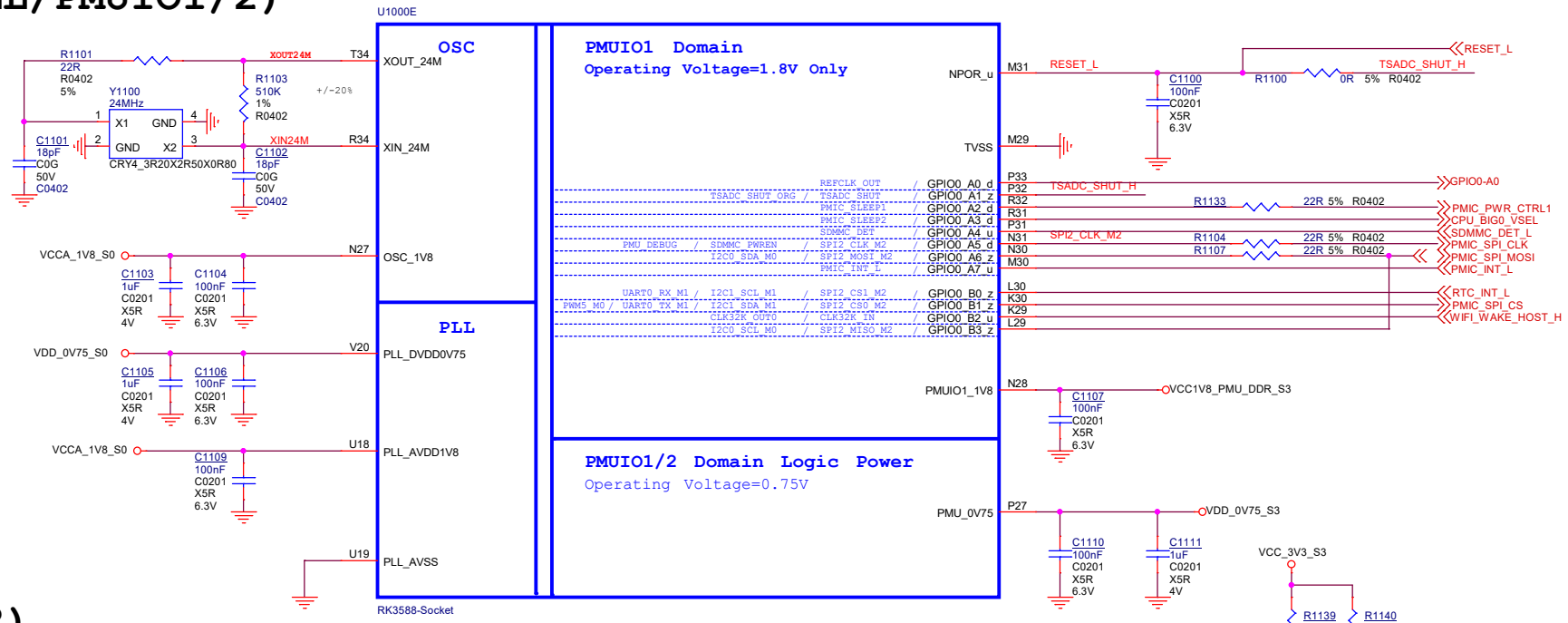
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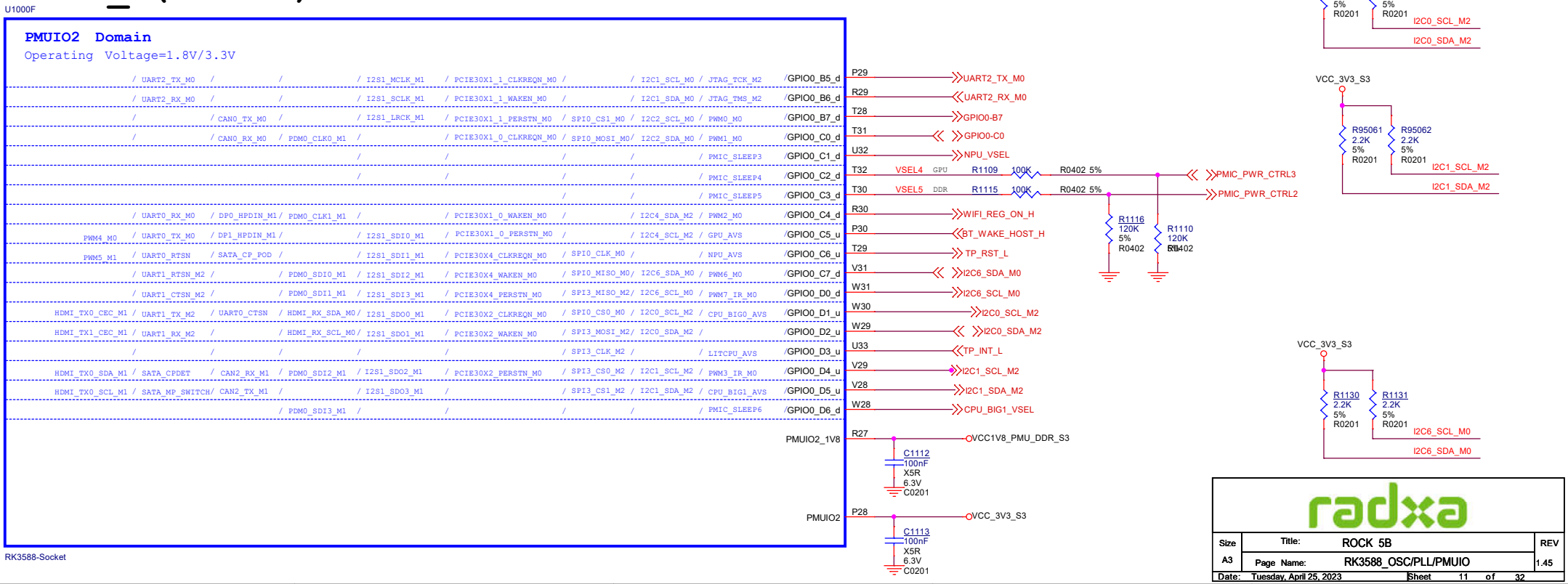


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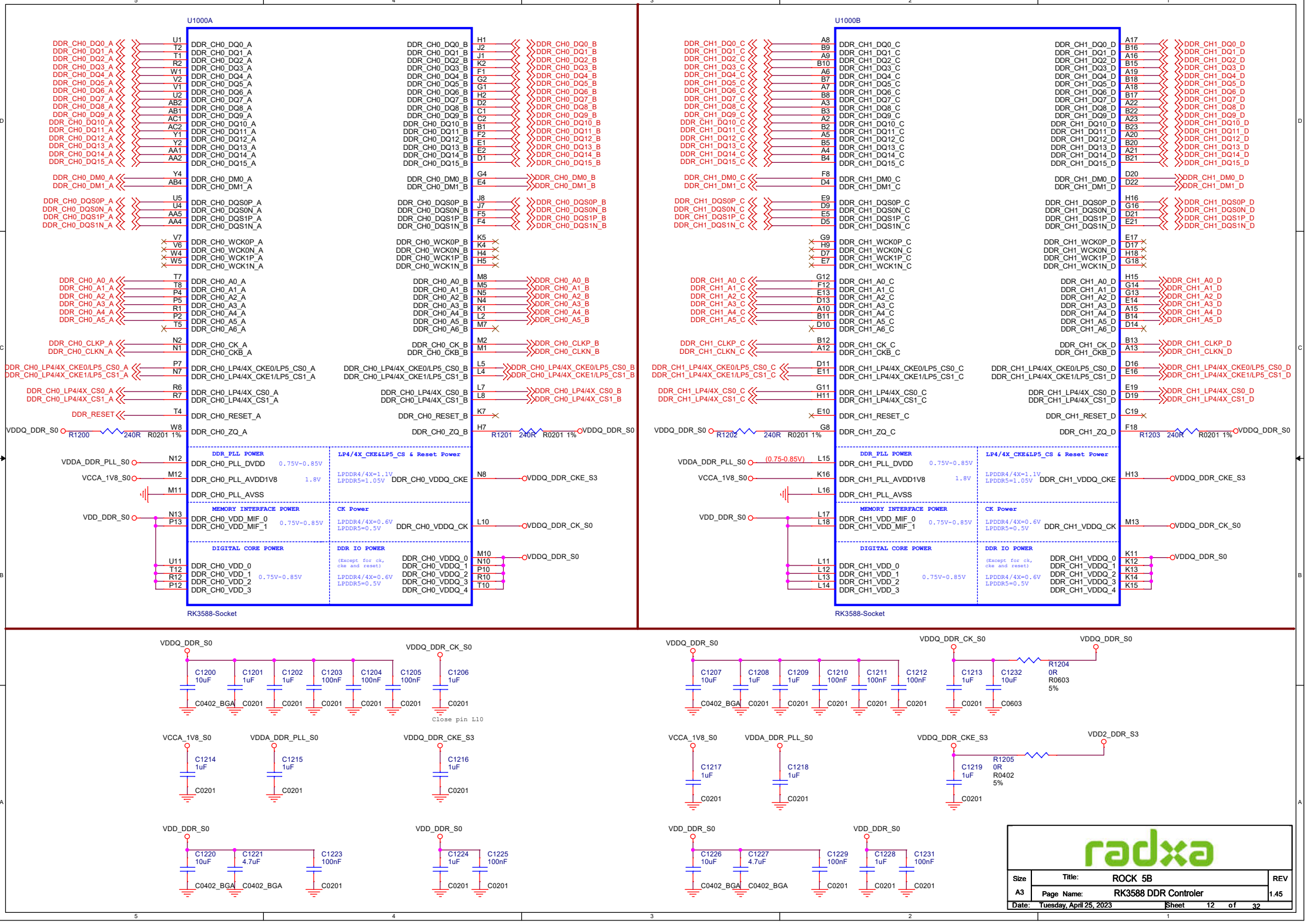
RK3588_E(OSC/PLL/PMUIO1/2)



RK3588_F(PMUIO2)



Size A3	Title: ROCK 5B		REV 1.45
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U1000A
DDR_CH0_DQ0_A
DDR_CH0_DQ1_A
DDR_CH0_DQ2_A
DDR_CH0_DQ3_A
DDR_CH0_DQ4_A
DDR_CH0_DQ5_A
DDR_CH0_DQ6_A
DDR_CH0_DQ7_A
DDR_CH0_DQ8_A
DDR_CH0_DQ9_A
DDR_CH0_DQ10_A
DDR_CH0_DQ11_A
DDR_CH0_DQ12_A
DDR_CH0_DQ13_A
DDR_CH0_DQ14_A
DDR_CH0_DQ15_A
DDR_CH0_DM0_A
DDR_CH0_DM1_A
DDR_CH0_DQS0P_A
DDR_CH0_DQS0N_A
DDR_CH0_DQS1P_A
DDR_CH0_DQS1N_A
DDR_CH0_WCK0P_A
DDR_CH0_WCK0N_A
DDR_CH0_WCK1P_A
DDR_CH0_WCK1N_A
DDR_CH0_A0_A
DDR_CH0_A1_A
DDR_CH0_A2_A
DDR_CH0_A3_A
DDR_CH0_A4_A
DDR_CH0_A5_A
DDR_CH0_A6_A
DDR_CH0_CK_A
DDR_CH0_CKB_A
DDR_CH0_LP4/4X_CKE0/LP5_CS0_A
DDR_CH0_LP4/4X_CKE1/LP5_CS1_A
DDR_CH0_LP4/4X_CS0_A
DDR_CH0_LP4/4X_CS1_A
DDR_CH0_RESET_A
DDR_CH0_ZQ_A
DDR_CH0_DQ0_B
DDR_CH0_DQ1_B
DDR_CH0_DQ2_B
DDR_CH0_DQ3_B
DDR_CH0_DQ4_B
DDR_CH0_DQ5_B
DDR_CH0_DQ6_B
DDR_CH0_DQ7_B
DDR_CH0_DQ8_B
DDR_CH0_DQ9_B
DDR_CH0_DQ10_B
DDR_CH0_DQ11_B
DDR_CH0_DQ12_B
DDR_CH0_DQ13_B
DDR_CH0_DQ14_B
DDR_CH0_DQ15_B
DDR_CH0_DM0_B
DDR_CH0_DM1_B
DDR_CH0_DQS0P_B
DDR_CH0_DQS0N_B
DDR_CH0_DQS1P_B
DDR_CH0_DQS1N_B
DDR_CH0_WCK0P_B
DDR_CH0_WCK0N_B
DDR_CH0_WCK1P_B
DDR_CH0_WCK1N_B
DDR_CH0_A0_B
DDR_CH0_A1_B
DDR_CH0_A2_B
DDR_CH0_A3_B
DDR_CH0_A4_B
DDR_CH0_A5_B
DDR_CH0_CK_B
DDR_CH0_CKB_B
DDR_CH0_LP4/4X_CKE0/LP5_CS0_B
DDR_CH0_LP4/4X_CKE1/LP5_CS1_B
DDR_CH0_LP4/4X_CS0_B
DDR_CH0_LP4/4X_CS1_B
DDR_CH0_RESET_B
DDR_CH0_ZQ_B
VDDQ_DDR_S0
R1200 240R R0201 1%
R1201 240R R0201 1%
DDR_PLL POWER
DDR_CH0_PLL_DVDD 0.75V~0.85V
VDDA_DDR_PLL_S0
DDR_CH0_PLL_AVDD1V8 1.8V
VCCA_1V8_S0
DDR_CH0_PLL_AVSS
MEMORY INTERFACE POWER
DDR_CH0_VDD_MIF_0
DDR_CH0_VDD_MIF_1 0.75V~0.85V
VDD_DDR_S0
DIGITAL CORE POWER
DDR_CH0_VDD_0
DDR_CH0_VDD_1 0.75V~0.85V
DDR_CH0_VDD_2
DDR_CH0_VDD_3
LP4/4X_CKE&LP5_CS & Reset Power
LPDDR4/4X=1.1V
LPDDR5=1.05V
DDR_CH0_VDDQ_CKE
VDDQ_DDR_CKE_S3
CK Power
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH0_VDDQ_CK
VDDQ_DDR_CK_S0
DDR IO POWER
(Except for ck, cke and reset)
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH0_VDDQ_0
DDR_CH0_VDDQ_1
DDR_CH0_VDDQ_2
DDR_CH0_VDDQ_3
DDR_CH0_VDDQ_4
VDDQ_DDR_S0
RK3588-Socket
U1000B
DDR_CH1_DQ0_C
DDR_CH1_DQ1_C
DDR_CH1_DQ2_C
DDR_CH1_DQ3_C
DDR_CH1_DQ4_C
DDR_CH1_DQ5_C
DDR_CH1_DQ6_C
DDR_CH1_DQ7_C
DDR_CH1_DQ8_C
DDR_CH1_DQ9_C
DDR_CH1_DQ10_C
DDR_CH1_DQ11_C
DDR_CH1_DQ12_C
DDR_CH1_DQ13_C
DDR_CH1_DQ14_C
DDR_CH1_DQ15_C
DDR_CH1_DM0_C
DDR_CH1_DM1_C
DDR_CH1_DQS0P_C
DDR_CH1_DQS0N_C
DDR_CH1_DQS1P_C
DDR_CH1_DQS1N_C
DDR_CH1_WCK0P_C
DDR_CH1_WCK0N_C
DDR_CH1_WCK1P_C
DDR_CH1_WCK1N_C
DDR_CH1_A0_C
DDR_CH1_A1_C
DDR_CH1_A2_C
DDR_CH1_A3_C
DDR_CH1_A4_C
DDR_CH1_A5_C
DDR_CH1_A6_C
DDR_CH1_CK_C
DDR_CH1_CKB_C
DDR_CH1_LP4/4X_CKE0/LP5_CS0_C
DDR_CH1_LP4/4X_CKE1/LP5_CS1_C
DDR_CH1_LP4/4X_CS0_C
DDR_CH1_LP4/4X_CS1_C
DDR_CH1_RESET_C
DDR_CH1_ZQ_C
DDR_CH1_DQ0_D
DDR_CH1_DQ1_D
DDR_CH1_DQ2_D
DDR_CH1_DQ3_D
DDR_CH1_DQ4_D
DDR_CH1_DQ5_D
DDR_CH1_DQ6_D
DDR_CH1_DQ7_D
DDR_CH1_DQ8_D
DDR_CH1_DQ9_D
DDR_CH1_DQ10_D
DDR_CH1_DQ11_D
DDR_CH1_DQ12_D
DDR_CH1_DQ13_D
DDR_CH1_DQ14_D
DDR_CH1_DQ15_D
DDR_CH1_DM0_D
DDR_CH1_DM1_D
DDR_CH1_DQS0P_D
DDR_CH1_DQS0N_D
DDR_CH1_DQS1P_D
DDR_CH1_DQS1N_D
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DDR_CH1_WCK0N_D
DDR_CH1_WCK1P_D
DDR_CH1_WCK1N_D
DDR_CH1_A0_D
DDR_CH1_A1_D
DDR_CH1_A2_D
DDR_CH1_A3_D
DDR_CH1_A4_D
DDR_CH1_A5_D
DDR_CH1_A6_D
DDR_CH1_CK_D
DDR_CH1_CKB_D
DDR_CH1_LP4/4X_CKE0/LP5_CS0_D
DDR_CH1_LP4/4X_CKE1/LP5_CS1_D
DDR_CH1_LP4/4X_CS0_D
DDR_CH1_LP4/4X_CS1_D
DDR_CH1_RESET_D
DDR_CH1_ZQ_D
VDDQ_DDR_S0
R1202 240R R0201 1%
R1203 240R R0201 1%
DDR_PLL POWER
DDR_CH1_PLL_DVDD 0.75V~0.85V
VDDA_DDR_PLL_S0 (0.75-0.85V)
DDR_CH1_PLL_AVDD1V8 1.8V
VCCA_1V8_S0
DDR_CH1_PLL_AVSS
MEMORY INTERFACE POWER
DDR_CH1_VDD_MIF_0
DDR_CH1_VDD_MIF_1 0.75V~0.85V
VDD_DDR_S0
DIGITAL CORE POWER
DDR_CH1_VDD_0
DDR_CH1_VDD_1 0.75V~0.85V
DDR_CH1_VDD_2
DDR_CH1_VDD_3
LP4/4X_CKE&LP5_CS & Reset Power
LPDDR4/4X=1.1V
LPDDR5=1.05V
DDR_CH1_VDDQ_CKE
VDDQ_DDR_CKE_S3
CK Power
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH1_VDDQ_CK
VDDQ_DDR_CK_S0
DDR IO POWER
(Except for ck, cke and reset)
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH1_VDDQ_0
DDR_CH1_VDDQ_1
DDR_CH1_VDDQ_2
DDR_CH1_VDDQ_3
DDR_CH1_VDDQ_4
VDDQ_DDR_S0
RK3588-Socket
VDDQ_DDR_S0
C1200 10uF
C1201 1uF
C1202 1uF
C1203 100nF
C1204 100nF
C1205 100nF
C1206 1uF
C0402_BGA
C0201
Close pin L10
VDDQ_DDR_CK_S0
VCCA_1V8_S0
C1214 1uF
VDDA_DDR_PLL_S0
C1215 1uF
VDDQ_DDR_CKE_S3
C1216 1uF
VDD_DDR_S0
C1220 10uF
C1221 4.7uF
C1223 100nF
C1224 1uF
C1225 100nF
C1207 10uF
C1208 1uF
C1209 1uF
C1210 100nF
C1211 100nF
C1212 100nF
C1213 1uF
C1232 10uF
C0603
R1204 0R R0603 5%
C1217 1uF
C1218 1uF
C1219 1uF
R1205 0R R0402 5%
VDD2_DDR_S3
C1226 10uF
C1227 4.7uF
C1229 100nF
C1228 1uF
C1231 100nF
radxa
Size A3
Title: ROCK 5B
REV 1.45
Page Name: RK3588 DDR Controler
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RK3588_C(EMMCIO Domain)

U1000C

EMMCIO Domain

Operating Voltage=1.8V

FSPI_D0_M0 / EMMC_D0	GPIO2_D0_u	Y33	R1306	0R	eMMC_D0
FSPI_D1_M0 / EMMC_D1	GPIO2_D1_u	W33	R1307	0R	eMMC_D1
FSPI_D2_M0 / EMMC_D2	GPIO2_D2_u	V32	R1308	0R	eMMC_D2
FSPI_D3_M0 / EMMC_D3	GPIO2_D3_u	AA33	R1309	0R	eMMC_D3
UART5_RX_M2 / I2C1_SCL_M3 / EMMC_D4	GPIO2_D4_u	Y32	R1314	0R	eMMC_D4
UART5_TX_M2 / I2C1_SDA_M3 / EMMC_D5	GPIO2_D5_u	AA32	R1315	0R	eMMC_D5
FSPI_CS0N_M0 / EMMC_D6	GPIO2_D6_u	W32	R1311	0R	eMMC_D6
FSPI_CS1N_M0 / EMMC_D7	GPIO2_D7_u	V33	R1316	0R	eMMC_D7
FSPI_CLK_M0 / EMMC_CMD	GPIO2_A0_u	W34	R1310	0R	eMMC_CMD
/ EMMC_CLKOUT	GPIO2_A1_d	V34	R1317	0R	eMMC_CLKOUT
UART5_CTSN_M1 / I2C2_SDA_M2 / EMMC_DATA_STROBE	GPIO2_A2_d	Y34			eMMC_DATA_STROBE
UART5_RTSN_M1 / I2C2_SCL_M2 / EMMC_RSTN	GPIO2_A3_d	AA34	R1318	0R	eMMC_RSTn

EMMCIO_1V8 — V26 — VCC_1V8_S3

C1300 1uF X5R 4V C0201

RK3588-Socket

RK3588_D(VCCIO2 Domain)

U1000D

VCCIO2 Domain

Operating Voltage=1.8V/3.3V

PWM8_M1 / I2C3_SCL_M4 / PDM1_SDI3_M0 / JTAG_TCK_M1 / UART2_TX_M1 / SDMMC_D0	GPIO4_D0_u	AD2	SDMMC0_D0
PWM9_M1 / I2C3_SDA_M4 / PDM1_SDI2_M0 / JTAG_TMS_M1 / UART2_RX_M1 / SDMMC_D1	GPIO4_D1_u	AD1	SDMMC0_D1
/ I2C8_SCL_M0 / PDM1_SDI1_M0 / JTAG_TCK_M0 / UART5_CTSN_M0 / SDMMC_D2	GPIO4_D2_u	AF2	SDMMC_D2/JTAG_TCK_M0
PWM10_M1 / I2C8_SDA_M0 / PDM1_SDI0_M0 / JTAG_TMS_M0 / UART5_RTSN_M0 / SDMMC_D3	GPIO4_D3_u	AF1	SDMMC_D3/JTAG_TMS_M0
PWM7_IR_M1 / CAN0_TX_M1 / PDM1_CLK1_M0 / MCU_JTAG_TCK_M0 / UART5_RX_M0 / SDMMC_CMD	GPIO4_D4_u	AE2	SDMMC_CMD/MCU_JTAG_TCK_M0
TEST_CLKOUT_M0 / CAN0_RX_M1 / PDM1_CLK0_M0 / MCU_JTAG_TMS_M0 / UART5_TX_M0 / SDMMC_CLK	GPIO4_D5_d	AE1	SDMMC_CLK/MCU_JTAG_TMS_M0

VCCIO2_1V8 — AA7 — VCC_1V8_S0

C1301 100nF C0201 X5R 6.3V

VCCIO2 — Y7 — VCCIO_SD_S0

C1302 100nF C0201 X5R 6.3V

RK3588-Socket

radxa

Size A4	Title: ROCK 5B	REV 1.45
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RK3588_M(TYPEC/DP)

U1000M

USB3.0 OTG/DP1.4 Alt of TYPEC0

USB:U3/Gen1----Controller0
DP:RBR/HBR/HBR2/HBR3

TYPEC0_SBU1/DP0_AUXP — AL15 — TYPEC0_SBU1
TYPEC0_SBU2/DP0_AUXN — AM15 — TYPEC0_SBU2
TYPEC0_SSRX1P/DP0_TX0P — AN13 — TYPEC0_SSRX1P
TYPEC0_SSRX1N/DP0_TX0N — AP13 — TYPEC0_SSRX1N
TYPEC0_SSTX1P/DP0_TX1P — AP14 — TYPEC0_SSTX1P
TYPEC0_SSTX1N/DP0_TX1N — AN14 — TYPEC0_SSTX1N
TYPEC0_SSRX2P/DP0_TX2P — AN15 — TYPEC0_SSRX2P
TYPEC0_SSRX2N/DP0_TX2N — AP15 — TYPEC0_SSRX2N
TYPEC0_SSTX2P/DP0_TX3P — AP16 — TYPEC0_SSTX2P
TYPEC0_SSTX2N/DP0_TX3N — AN16 — TYPEC0_SSTX2N
TYPEC0_DP0_REXT — AH16 — TYPEC0_REXT — R1402 8.2K R0201 1%
TYPEC0_DP0_VDD_0V85 — AJ14
TYPEC0_DP0_VDDA_0V85 — AH14 — VDDA_0V85_S0
C1400 1uF X5R 6.3V C0201
C1401 100nF X5R 6.3V C0201
TYPEC0_DP0_VDDH_1V8 — AG14 — VCCA_1V8_S0
C1402 1uF X5R 6.3V C0201
C1403 100nF X5R 6.3V C0201

USB3.0 OTG/DP1.4 Alt of TYPEC1

USB:U3/Gen1----Controller1
DP:RBR/HBR/HBR2/HBR3

TYPEC1_SBU1/DP1_AUXP — AL10 — DP1_AUXP
TYPEC1_SBU2/DP1_AUXN — AM10 — DP1_AUXN
TYPEC1_SSRX1P/DP1_TX0P — AN8 — TYPEC1_SSRX1P
TYPEC1_SSRX1N/DP1_TX0N — AP8 — TYPEC1_SSRX1N
TYPEC1_SSTX1P/DP1_TX1P — AP9 — TYPEC1_SSTX1P
TYPEC1_SSTX1N/DP1_TX1N — AN9 — TYPEC1_SSTX1N
TYPEC1_SSRX2P/DP1_TX2P — AN10 — DP1_TX2P
TYPEC1_SSRX2N/DP1_TX2N — AP10 — DP1_TX2N
TYPEC1_SSTX2P/DP1_TX3P — AP11 — DP1_TX3P
TYPEC1_SSTX2N/DP1_TX3N — AN11 — DP1_TX3N
TYPEC1_DP1_REXT — AG16 — TYPEC1_REXT — R1410 8.2K R0201 1%
TYPEC1_DP1_VDD_0V85 — AH13
TYPEC1_DP1_VDDA_0V85 — AJ13 — VDDA_0V85_S0
C1404 1uF X5R 6.3V C0201
C1405 100nF X5R 6.3V C0201
TYPEC1_DP1_VDDH_1V8 — AG13 — VCCA_1V8_S0
C1406 1uF X5R 6.3V C0201
C1407 100nF X5R 6.3V C0201

RK3588-Socket

RK3588_L(USB2.0 HOST/OTG)

U1000L

USB2.0 of TYPEC0 (OTG/HOST/DEVICE)
HS/FS/LS

Download Port
TYPEC0_USB20_OTG_DP — AL12 — TYPEC0_OTG_DP
TYPEC0_USB20_OTG_DM — AM12 — TYPEC0_OTG_DM
TYPEC0_USB20_OTG_ID — AL14 — TYPEC0_USB20_OTG_ID
40K TYPEC0_USB20_VBUSDET — AM14 — TYPEC0_USB20_VBUSDET
TYPEC0_USB20_OTG0_REXT — AP12 — OTG0_REXT — R1405 200R R0201 1%

USB2.0 of TYPEC1 (OTG/HOST/DEVICE)
HS/FS/LS

TYPEC1_USB20_OTG_DP — AK9 — TYPEC1_OTG_DP
TYPEC1_USB20_OTG_DM — AL9 — TYPEC1_OTG_DM
TYPEC1_USB20_OTG_ID — AK8
40K TYPEC1_USB20_VBUSDET — AL8
TYPEC1_USB20_OTG1_REXT — AP7 — OTG1_REXT — R1406 200R R0201 1%

USB2.0 HOST0
HS/FS/LS

USB20_HOST0_DP — AK6 — USB20_HOST0_DP
USB20_HOST0_DM — AL6 — USB20_HOST0_DM
USB20_HOST0_REXT — AG9 — HOST0_REXT — R1400 200R R0201 1%

USB2.0 HOST1
HS/FS/LS

USB20_HOST1_DP — AL7 — USB20_HOST1_DP
USB20_HOST1_DM — AM7 — USB20_HOST1_DM
USB20_HOST1_REXT — AH9 — HOST1_REXT — R1401 200R R0201 1%

USB2.0 POWER

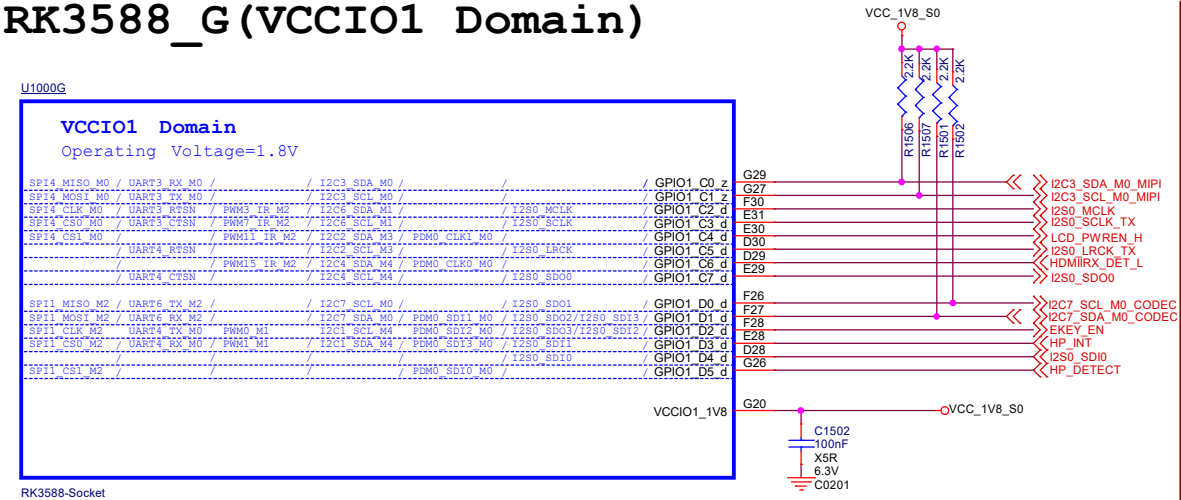
USB20_DVDD_0V75 — AH10 — VDDA_0V75_S0
C1408 100nF X5R 6.3V C0201
USB20_AVDD_1V8 — AG11 — VCCA_1V8_S0
C1409 100nF X5R 6.3V C0201
USB20_AVDD_3V3 — AJ10 — VCCA_3V3_S0
C1410 100nF X5R 6.3V C0201

RK3588-Socket

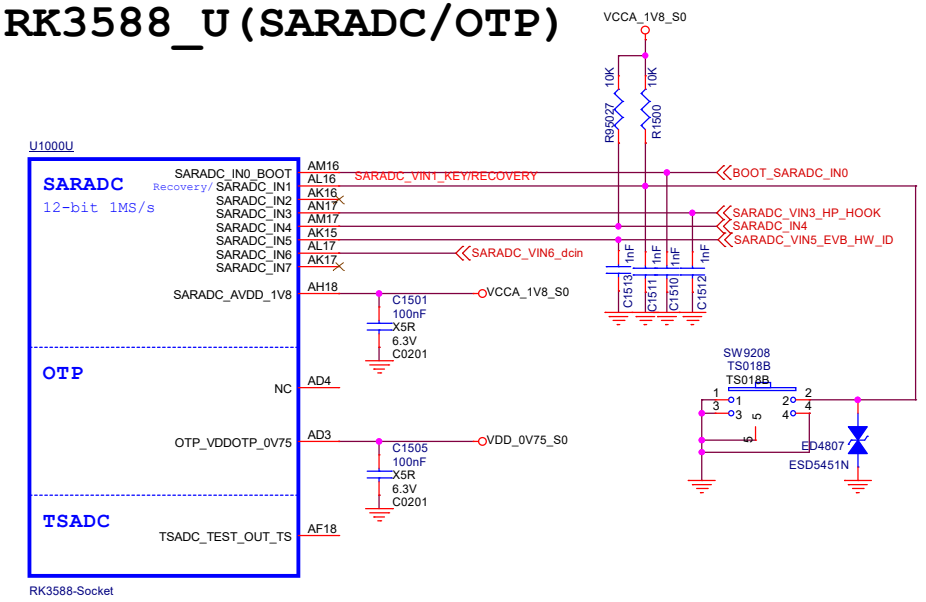
radxa

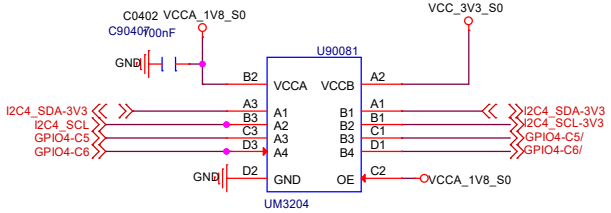
Size	Title:	REV
A3	ROCK 5B	
	Page Name: RK3588_USB30/USB20_Ctrl	1.45
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RK3588_G(VCCIO1 Domain)

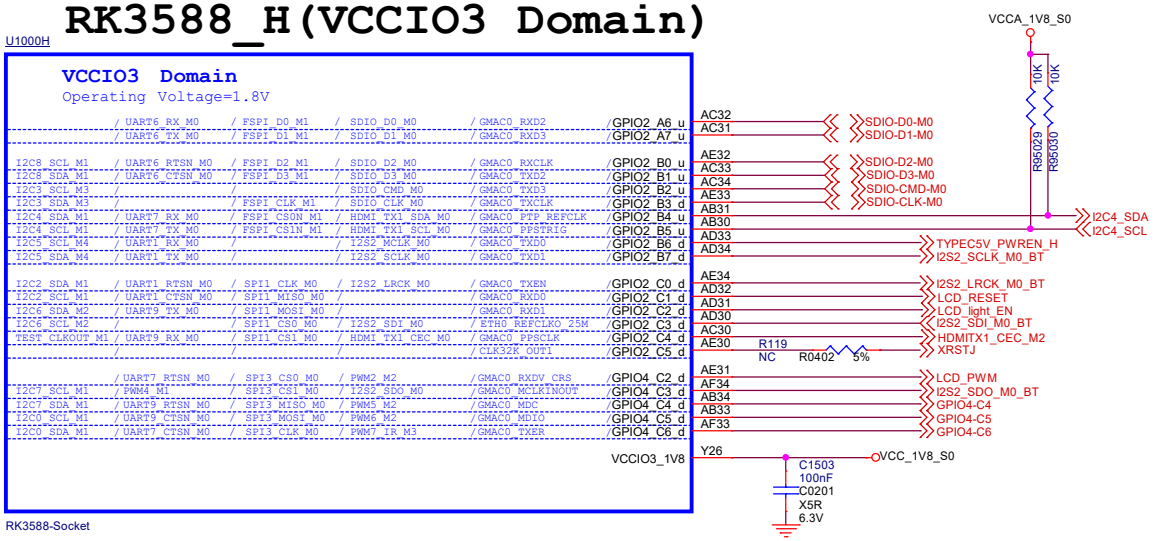


RK3588_U(SARADC/OTP)





RK3588_H(VCCIO3 Domain)



BOOT MODE CONFIG

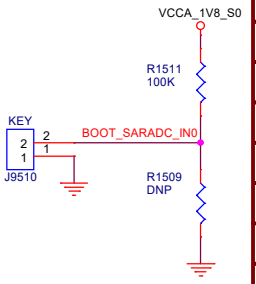


TABLE 1

Item	Rup	Rdown	ADC	VOL	BOOT MODE
LEVEL1	DNP	100K	0	0V	USB (Maskrom mode)
LEVEL2	100K	20K	682	0.3V	SD Card-USB
LEVEL3	100K	51K	1365	0.6V	EMMC-USB
LEVEL4	100K	100K	2047	0.9V	FSPI M0-USB
LEVEL5	100K	200K	2730	1.2V	FSPI M1-USB
LEVEL6	100K	499K	3412	1.5V	FSPI M2-USB
LEVEL7	100K	DNP	4095	1.8V	FSPI_M2-FSPI_M1-FSPI_M0 -EMMC-SD Card-USB

BOARD ID CONFIG

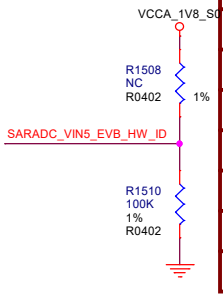


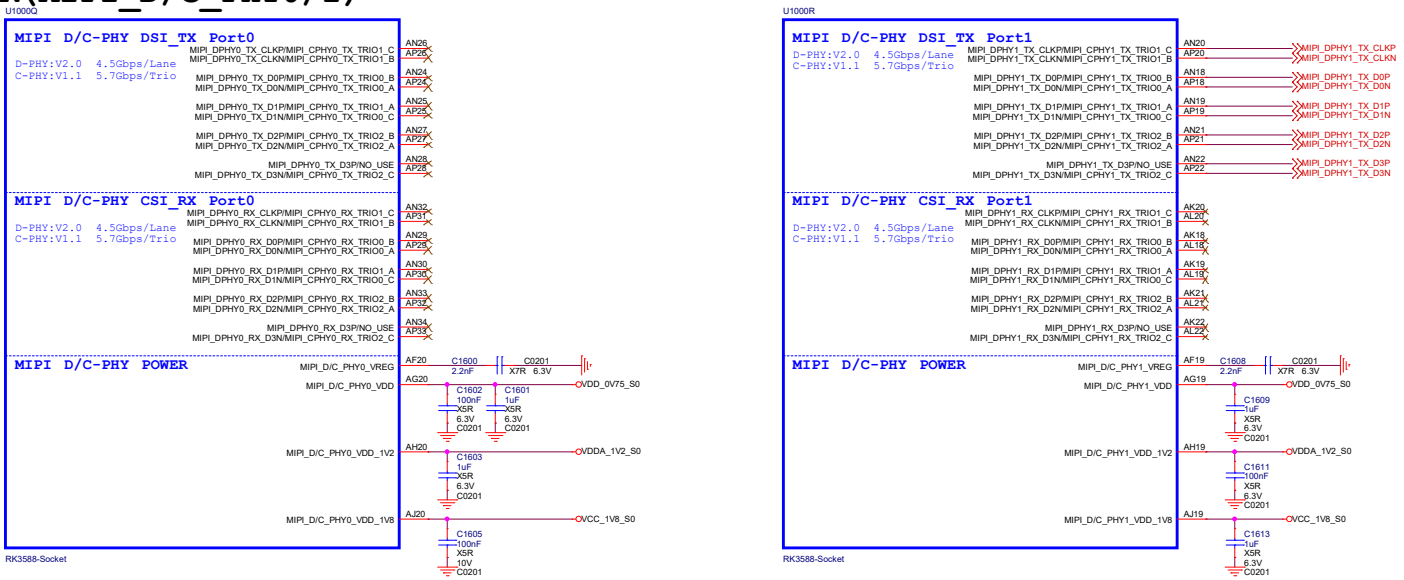
TABLE 2

Item	Rup	Rdown	ADC	VOL	VERSION
LEVEL1	DNP	100K	0	0V	A
LEVEL2	100K	20K	682	0.3V	B
LEVEL3	100K	51K	1365	0.6V	C
LEVEL4	100K	100K	2047	0.9V	D
LEVEL5	100K	200K	2730	1.2V	E
LEVEL6	100K	499K	3412	1.5V	F
LEVEL7	100K	DNP	4095	1.8V	H

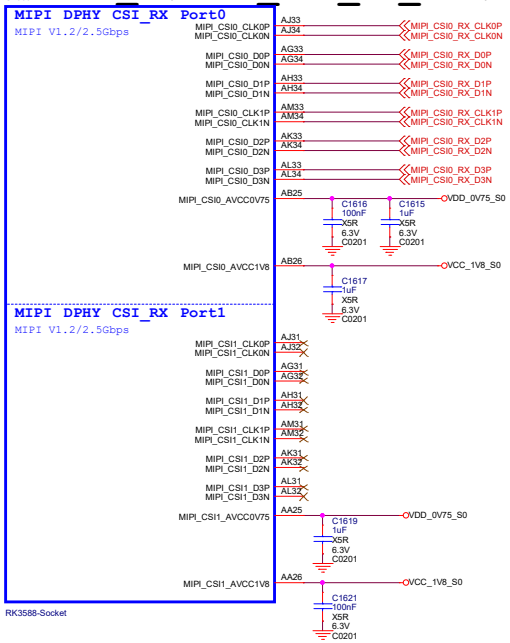
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Size	Title:	REV
A3	ROCK 5B	
	Page Name: RK3588_SARADC/1.8V Only GPIO	1.45
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RK3588_Q/R(MIPI_D/C_PHY0/1)



RK3588_P(MIPI_CSI_RX_PHY)



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Size	Title: ROCK 5B	REV
A2	Page Name: RK3588_MIPI Interface	1.45
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RK3588_S(HDMI2.1 TX)

U1000S

HDMI TX/eDP MUX Port0

HDMI:V2.1 12Gbps
eDP: V1.3 5.4Gbps

HDMI_TX0_SBDP/EDP_TX0_AUXP	AG2	HDMI0_TX_SBDP/eDP0_TX_AUXP	HDMI0_TX_SBDP/eDP0_TX_AUXP
HDMI_TX0_SBDN/EDP_TX0_AUXN	AG1	HDMI0_TX_SBDN/eDP0_TX_AUXN	HDMI0_TX_SBDN/eDP0_TX_AUXN
HDMI_TX0_D0P/EDP_TX0_D0P	AJ2	HDMI0_TX0P_PORT/eDP0_TX_D0P	HDMI0_TX0P_PORT/eDP0_TX_D0P
HDMI_TX0_D0N/EDP_TX0_D0N	AJ1	HDMI0_TX0N_PORT/eDP0_TX_D0N	HDMI0_TX0N_PORT/eDP0_TX_D0N
HDMI_TX0_D1P/EDP_TX0_D1P	AK3	HDMI0_TX1P_PORT/eDP0_TX_D1P	HDMI0_TX1P_PORT/eDP0_TX_D1P
HDMI_TX0_D1N/EDP_TX0_D1N	AK2	HDMI0_TX1N_PORT/eDP0_TX_D1N	HDMI0_TX1N_PORT/eDP0_TX_D1N
HDMI_TX0_D2P/EDP_TX0_D2P	AL2	HDMI0_TX2P_PORT/eDP0_TX_D2P	HDMI0_TX2P_PORT/eDP0_TX_D2P
HDMI_TX0_D2N/EDP_TX0_D2N	AL1	HDMI0_TX2N_PORT/eDP0_TX_D2N	HDMI0_TX2N_PORT/eDP0_TX_D2N
HDMI_TX0_D3P/EDP_TX0_D3P	AH3	HDMI0_TX3P_PORT/eDP0_TX_D3P	HDMI0_TX3P_PORT/eDP0_TX_D3P
HDMI_TX0_D3N/EDP_TX0_D3N	AH2	HDMI0_TX3N_PORT/eDP0_TX_D3N	HDMI0_TX3N_PORT/eDP0_TX_D3N

HDMI/EDP_TX0_REXT — AM2 HDMI_TX0_REXT — R1708 8.2K R0201 1%

HDMI/EDP_TX0_VDD_0V75 — AA9 <260mA — VDDA_0V75_S0
C1715 4.7uF X5R 10V C0402_BGA
C1751 4.7uF X5R 10V C0402
VDD_IO<77mA
VDD_CMN<71mA

HDMI/EDP_TX0_AVDD_0V75 — AB9 <1mA
C1714 1uF X5R 6.3V C0201

HDMI/EDP_TX0_VDD_IO_1V8 — AC7
HDMI/EDP_TX0_VDD_CMN_1V8 — AC6 — VCC_1V8_S0
C1716 1uF X5R 6.3V C0201
C1717 100nF X5R 6.3V C0201
C1742 4.7uF X5R 10V C0402

HDMI TX/eDP MUX Port1

HDMI:V2.1 12Gbps
eDP: V1.3 5.4Gbps

HDMI_TX1_SBDP/EDP_TX1_AUXP	AN2	HDMI1_TX_SBDP/eDP1_TX_AUXP	HDMI1_TX_SBDP/eDP1_TX_AUXP
HDMI_TX1_SBDN/EDP_TX1_AUXN	AP2	HDMI1_TX_SBDN/eDP1_TX_AUXN	HDMI1_TX_SBDN/eDP1_TX_AUXN
HDMI_TX1_D0P/EDP_TX1_D0P	AN4	HDMI1_TX0P_PORT/eDP1_TX_D0P	HDMI1_TX0P_PORT/eDP1_TX_D0P
HDMI_TX1_D0N/EDP_TX1_D0N	AP4	HDMI1_TX0N_PORT/eDP1_TX_D0N	HDMI1_TX0N_PORT/eDP1_TX_D0N
HDMI_TX1_D1P/EDP_TX1_D1P	AM5	HDMI1_TX1P_PORT/eDP1_TX_D1P	HDMI1_TX1P_PORT/eDP1_TX_D1P
HDMI_TX1_D1N/EDP_TX1_D1N	AN5	HDMI1_TX1N_PORT/eDP1_TX_D1N	HDMI1_TX1N_PORT/eDP1_TX_D1N
HDMI_TX1_D2P/EDP_TX1_D2P	AN6	HDMI1_TX2P_PORT/eDP1_TX_D2P	HDMI1_TX2P_PORT/eDP1_TX_D2P
HDMI_TX1_D2N/EDP_TX1_D2N	AP6	HDMI1_TX2N_PORT/eDP1_TX_D2N	HDMI1_TX2N_PORT/eDP1_TX_D2N
HDMI_TX1_D3P/EDP_TX1_D3P	AM3	HDMI1_TX3P_PORT/eDP1_TX_D3P	HDMI1_TX3P_PORT/eDP1_TX_D3P
HDMI_TX1_D3N/EDP_TX1_D3N	AN3	HDMI1_TX3N_PORT/eDP1_TX_D3N	HDMI1_TX3N_PORT/eDP1_TX_D3N

HDMI/EDP_TX1_REXT — AN1 HDMI_TX1_REXT — R1719 8.2K 1% R0201

HDMI/EDP_TX1_VDD_0V75 — AD9 <260mA — VDDA_0V75_S0
C1735 1uF X5R 6.3V C0201
C1752 100nF X5R 10V C0201
C1730 4.7uF X5R 10V C0402

HDMI/EDP_TX1_AVDD_0V75 — AC9 <1mA
C1729 1uF X5R 6.3V C0201

HDMI/EDP_TX1_VDD_IO_1V8 — AD7
HDMI/EDP_TX1_VDD_CMN_1V8 — AD6 — VCC_1V8_S0
C1731 1uF X5R 6.3V C0201
C1732 100nF X5R 6.3V C0201
C1733 4.7uF X5R 10V C0402

RK3588-Socket

RK3588_T(HDMI20 RX)

U1000T

HDMI RX

HDMI:V2.0

HDMI_RX_D0P	AG5	HDMI_RX_D0P	HDMI_RX_D0P_PORT
HDMI_RX_D0N	AG4	HDMI_RX_D0N	HDMI_RX_D0N_PORT
HDMI_RX_D1P	AH6	HDMI_RX_D1P	HDMI_RX_D1P_PORT
HDMI_RX_D1N	AH5	HDMI_RX_D1N	HDMI_RX_D1N_PORT
HDMI_RX_D2P	AJ5	HDMI_RX_D2P	HDMI_RX_D2P_PORT
HDMI_RX_D2N	AJ4	HDMI_RX_D2N	HDMI_RX_D2N_PORT
HDMI_RX_CLKP	AF6	HDMI_RX_CLKP	HDMI_RX_CLKP_PORT
HDMI_RX_CLKN	AF5	HDMI_RX_CLKN	HDMI_RX_CLKN_PORT

HDMI_RX_REXT — AF3 — R1730 200R R0201 1%

HDMI_RX_AVDD0V75 — AE8 — VDDA_0V75_S0 <40mA
C1738 1uF X5R 6.3V C0201

HDMI_RX_VPH3V3 — AE4
HDMI_RX_DVDD3V3 — AE5 — VCCA_3V3_S0 <80mA
C1740 100nF X5R 6.3V C0201
C1741 1uF X5R 6.3V C0201

RK3588-Socket

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RK3588_N(PCIE20)

U1000N

PCIE20/SATA30 Mux0

PCIe20x1_2 1Lane(RC)--4(1L2)

SATA30 HOST Controller0

MUX

input or output

PCIE20_0_REFCLKP L32 → PCIE20_0_REFCLKP
PCIE20_0_REFCLKN L33 → PCIE20_0_REFCLKN
PCIE20_0_TXP/SATA30_0_TXP M34 → PCIE20_0_TXP
PCIE20_0_TXN/SATA30_0_TXN M33 → PCIE20_0_TXN
PCIE20_0_RXP/SATA30_0_RXP N33 → PCIE20_0_RXP
PCIE20_0_RXN/SATA30_0_RXN N34 → PCIE20_0_RXN
PCIE20_SATA30_0_AVDD_0V85 M28 → C1810 10uF X5R 4V C0402_BGA → VDDA_0V85_S0
PCIE20_SATA30_0_AVDD_1V8 M27 → C1808 1uF X5R 4V C0201 → VCCA_1V8_S0

PCIE20X1_2-CTL

PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3

PCIE20/SATA30 Mux1

PCIe30x1_0 1Lane(RC)--2(1L0)

SATA30 HOST Controller1

MUX

input or output

PCIE20_1_REFCLKP H32 → PCIE20_1_REFCLKP
PCIE20_1_REFCLKN H33 → PCIE20_1_REFCLKN
PCIE20_1_TXP/SATA30_1_TXP K33 → PCIE20_1_TXP
PCIE20_1_TXN/SATA30_1_TXN K34 → PCIE20_1_TXN
PCIE20_1_RXP/SATA30_1_RXP J33 → PCIE20_1_RXP
PCIE20_1_RXN/SATA30_1_RXN J34 → PCIE20_1_RXN
PCIE20_SATA30_1_AVDD_0V85 L28 → C1818 100nF X5R 6.3V C0201 → VDDA_0V85_S0
PCIE20_SATA30_1_AVDD_1V8 L27 → C1816 100nF X5R 6.3V C0201 → VCCA_1V8_S0

PCIE30X1_0-CTL

PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3

PCIE20/SATA30/USB30 HOST Mux2

PCIe30x1_1 1Lane(RC)--3(1L1)

SATA30 HOST Controller2

USB30 HOST Controller2

MUX

input or output

PCIE20_2_REFCLKP G31
PCIE20_2_REFCLKN G30
PCIE20_2_TXP/SATA30_2_TXP/USB30_2_SSTXP H30 → USB30_2_SSTXP
PCIE20_2_TXN/SATA30_2_TXN/USB30_2_SSTXN H29 → USB30_2_SSTXN
PCIE20_2_RXP/SATA30_2_RXP/USB30_2_SSRXP J31 → USB30_2_SSRXP
PCIE20_2_RXN/SATA30_2_RXN/USB30_2_SSRXN J30 → USB30_2_SSRXN
PCIE20_SATA30_USB30_2_AVDD_0V85 K28 → C1806 1uF X5R 4V C0201 → VDDA_0V85_S0
PCIE20_SATA30_USB30_2_AVDD_1V8 K27 → C1804 1uF X5R 4V C0201 → VCCA_1V8_S0

For RJ45

PCIE20:Gen1/Gen2
USB 30:Gen1
SATA30:Gen1/Gen2/Gen3

RK3588-Socket

RK3588_O(PCIE30)

U1000O

PCIE30X4

PCIe30X4 4Lane(DM)--0(4L)

PCIe30x1_0 1Lane(RC)--2(1L0)

PCIE20/SATA30 MUX1

PCIe30x2 2Lane(RC)--1(2L)

PCIe30x1_1 1Lane(RC)--3(1L1)

PCIE20/SATA30/USB30 MUX2

PCIE30:
PCIe1.1(2.5Gbps)
PCIe2.1(5Gbps)
PCIe3.0(8Gbps)

PCIE30 PORT0

input

PCIE30_PORT0_REF_CLKP E33 → PCIE30_PORT0_REFCLKP_IN
PCIE30_PORT0_REF_CLKN E34 → PCIE30_PORT0_REFCLKN_IN

Lane0

PCIE30_PORT0_TX0P D32 → PCIE30_PORT0_TX0P
PCIE30_PORT0_TX0N D33 → PCIE30_PORT0_TX0N
PCIE30_PORT0_RX0P G33 → PCIE30_PORT0_RX0P
PCIE30_PORT0_RX0N G34 → PCIE30_PORT0_RX0N

Lane1

PCIE30_PORT0_TX1P C33 → PCIE30_PORT0_TX1P
PCIE30_PORT0_TX1N C34 → PCIE30_PORT0_TX1N
PCIE30_PORT0_RX1P F32 → PCIE30_PORT0_RX1P
PCIE30_PORT0_RX1N F33 → PCIE30_PORT0_RX1N

PCIE30_PORT0_RESREF B34 → PCIE30_0_REF → R1801 200R 1% R0201

PCIE30_PORT0_AVDD0V75 G24 → C1802 1uF X5R 6.3V C0201, C1803 4.7uF X5R 10V C0402 → VDD_0V75_S0

PCIE30_PORT0_AVDD1V8 G23 → C1800 1uF X5R 6.3V C0201, C1801 4.7uF X5R 10V C0402 → VCCA_1V8_S0

PCIE30 PORT1

input

PCIE30_PORT1_REF_CLKP A28 → PCIE30_PORT1_REFCLKP_IN
PCIE30_PORT1_REF_CLKN B28 → PCIE30_PORT1_REFCLKN_IN

Lane0

PCIE30_PORT1_TX0P B30 → PCIE30_PORT1_TX2P
PCIE30_PORT1_TX0N A30 → PCIE30_PORT1_TX2N
PCIE30_PORT1_RX0P B32 → PCIE30_PORT1_RX2P
PCIE30_PORT1_RX0N A32 → PCIE30_PORT1_RX2N

Lane1

PCIE30_PORT1_TX1P C29 → PCIE30_PORT1_TX3P
PCIE30_PORT1_TX1N B29 → PCIE30_PORT1_TX3N
PCIE30_PORT1_RX1P C31 → PCIE30_PORT1_RX3P
PCIE30_PORT1_RX1N B31 → PCIE30_PORT1_RX3N

PCIE30_PORT1_RESREF A33 → PCIE30_1_REF → R1807 200R 1% R0201

PCIE30_PORT1_AVDD0V75 H24 → C1814 1uF X5R 6.3V C0201, C1815 4.7uF X5R C0402 10V → VDD_0V75_S0

PCIE30_PORT1_AVDD1V8 H23 → C1812 100nF X5R 6.3V C0201, C1813 4.7uF X5R 10V C0402 → VCCA_1V8_S0

RK3588-Socket

radxa

Size	Title:	REV
A3	ROCK 5B	1.45
	Page Name: RK3588_PCIE30/PCIE20/SATA30	
Date: Tuesday, April 25, 2023	Sheet 18 of 32	

RK3588_J(VCCIO5 Domain)

U1000J

VCCIO5 Domain

Operating Voltage=1.8V/3.3V

PWM10_M0	SPI4_MISO_M1		I2C6_SDA_M4	FSPI_D0_M2	I2S3_MCLK	SDIO_D0_M1	GMAC1_TXD2	GPIO3_A0_u	AA29
AUDDSM_LN	SPI4_MOSI_M1	PWM11_IR_M0	I2C6_SCL_M4	FSPI_D1_M2	I2S3_SCLK	SDIO_D1_M1	GMAC1_TXD3	GPIO3_A1_u	AA30
AUDDSM_LP	SPI4_CLK_M1	UART8_TX_M1		FSPI_D2_M2	I2S3_LRCK	SDIO_D2_M1	GMAC1_RXD2	GPIO3_A2_u	AD27
AUDDSM_RN	SPI4_CS0_M1	UART8_RX_M1		FSPI_D3_M2	I2S3_SDO	SDIO_D3_M1	GMAC1_RXD3	GPIO3_A3_u	AE27
AUDDSM_RP	SPI4_CS1_M1	UART8_RTSN_M1			I2S3_SDI	SDIO_CMD_M1	GMAC1_TXCLK	GPIO3_A4_d	AD28
	MIPI_CAMERA0_CLK_M1	UART8_CTSN_M1	I2C4_SDA_M0	FSPI_CLK_M2		SDIO_CLK_M1	GMAC1_RXCLK	GPIO3_A5_d	AH30
	MIPI_CAMERA1_CLK_M1		I2C4_SCL_M0				ETH1_REFCLKO_25M	GPIO3_A6_d	AH27
PWM8_M0	MIPI_CAMERA2_CLK_M1						GMAC1_RXD0	GPIO3_A7_u	AG29
PWM9_M0	MIPI_CAMERA3_CLK_M1						GMAC1_RXD1	GPIO3_B0_u	AG28
PWM2_M1	MIPI_CAMERA4_CLK_M1	UART2_TX_M2					GMAC1_RXDV_CRS	GPIO3_B1_d	AH29
PWM3_IR_M1		UART2_RX_M2			I2S2_SDI_M1		GMAC1_TXER	GPIO3_B2_d	AE28
		UART2_RTSN			I2S2_SDO_M1		GMAC1_TXD0	GPIO3_B3_u	AC28
		UART2_CTSN			I2S2_MCLK_M1		GMAC1_TXD1	GPIO3_B4_u	AC29
PWM12_M0	CAN1_RX_M0	UART3_TX_M1			I2S2_SCLK_M1		GMAC1_TXEN	GPIO3_B5_u	AD29
PWM13_M0	CAN1_TX_M0	UART3_RX_M1			I2S2_LRCK_M1		GMAC1_MCLKINOUT	GPIO3_B6_d	AE29
			I2C3_SCL_M1	SPI1_MOSI_M1	HDMI_TX1_HPD_M1		GMAC1_PTP_REF_CLK	GPIO3_B7_d	AA28
		UART7_TX_M1	I2C3_SDA_M1	SPI1_MISO_M1			GMAC1_PPSTRIG	GPIO3_C0_d	Y29
	PCIE30X2_BUTTON_RSTN	UART7_RX_M1		SPI1_CLK_M1			GMAC1_PPSCLK	GPIO3_C1_d	Y27
PWM14_M0		UART7_RTSN_M1	I2C8_SCL_M4	SPI1_CS0_M1		MIPI_TE0	GMAC1_MDC	GPIO3_C2_d	Y31
PWM15_IR_M0		UART7_CTSN_M1	I2C8_SDA_M4	SPI1_CS1_M1		MIPI_TE1	GMAC1_MDIO	GPIO3_C3_d	Y30
CAN2_RX_M0	PCIE30X4_CLKREQN_M2	UART5_TX_M1	FSPI_CS0N_M2	SPI3_CS0_M3	HDMI_TX1_CEC_M2		CIF_D8	GPIO3_C4_u	AH26
CAN2_TX_M0	PCIE30X4_WAKEN_M2	UART5_RX_M1	FSPI_CS1N_M2	SPI3_CS1_M3	HDMI_TX1_SDA_M1		CIF_D9	GPIO3_C5_u	AH25
	PCIE30X4_PERSTN_M2			SPI3_MISO_M3	HDMI_TX1_SCL_M1		CIF_D10	GPIO3_C6_u	AG26
	PCIE20X1_2_CLKREQN_M0		I2C5_SCL_M0	SPI3_MOSI_M3	HDMI_TX0_SCL_M2		CIF_D11	GPIO3_C7_u	AJ24
PWM8_M2	PCIE20X1_2_WAKEN_M0	UART4_RX_M1	I2C5_SDA_M0	SPI3_CLK_M3	HDMI_TX0_SDA_M2		CIF_D12	GPIO3_D0_u	AH24
PWM9_M2	PCIE20X1_2_PERSTN_M0	UART4_TX_M1		SPI0_MISO_M3	HDMI_RX_CEC_M1		CIF_D13	GPIO3_D1_d	AG23
	PCIE30X2_CLKREQN_M2	UART9_RTSN_M2	I2C7_SCL_M2	SPI0_MOSI_M3	HDMI_RX_SCL_M1		CIF_D14	GPIO3_D2_d	AG25
PWM10_M2	PCIE30X2_WAKEN_M2	UART9_CTSN_M2	I2C7_SDA_M2	SPI0_CLK_M3	HDMI_RX_SDA_M1		CIF_D15	GPIO3_D3_d	AG24
	PCIE30X2_PERSTN_M2	UART9_RX_M2		SPI0_CS0_M3	HDMI_RX_HPDOUT_M1/ HDMI_TX0_HPD_M1/		MCU_JTAG_TCK_M1	GPIO3_D4_d	AA27
PWM11_IR_M3	PCIE30X4_BUTTON_RSTN	UART9_TX_M2		SPI0_CS1_M3	DP1_HPDIN_M0		MCU_JTAG_TMS_M1	GPIO3_D5_d	AB28

VCCIO5_1V8 — W25 — C1903 100nF X5R 6.3V C0201 — VCC_1V8_S0

VCCIO5 — W26 — C1902 100nF X5R 6.3V C0201 — VCC_3V3_S0

TP6802 TP_2.0

RK3588-Socket

RK3588_K(VCCIO6 Domain)

U1000K

VCCIO6 Domain

Operating Voltage=1.8V/3.3V

	SPI0_MISO_M1	UART9_RTSN_M1			I2S1_MCLK_M0	PCIE30X1_1_CLKREQN_M1	BT1120_D0	CIF_D0	GPIO4_A0_d	AK30
	SPI0_MOSI_M1	UART9_CTSN_M1			I2S1_SCLK_M0	PCIE30X1_1_WAKEN_M1	BT1120_D1	CIF_D1	GPIO4_A1_d	AL30
	SPI0_CLK_M1				I2S1_LRCK_M0	PCIE30X1_1_PERSTN_M1	BT1120_D2	CIF_D2	GPIO4_A2_d	AM29
		UART0_TX_M2				PCIE30X1_0_CLKREQN_M1	BT1120_D3	CIF_D3	GPIO4_A3_d	AL29
	SPI2_MISO_M1	UART0_RX_M2	I2C3_SCL_M2			PCIE30X1_0_WAKEN_M1	BT1120_D4	CIF_D4	GPIO4_A4_d	AL28
	SPI2_MOSI_M1	UART3_TX_M2	I2C3_SDA_M2		I2S1_SDI0_M0	PCIE30X1_0_PERSTN_M1	BT1120_D5	CIF_D5	GPIO4_A5_d	AK27
	SPI2_CLK_M1	UART3_RX_M2	I2C5_SCL_M2		I2S1_SDI1_M0	PCIE30X2_CLKREQN_M1	BT1120_D6	CIF_D6	GPIO4_A6_d	AL27
	SPI2_CS0_M1		I2C5_SDA_M2		I2S1_SDI2_M0	PCIE30X2_WAKEN_M1	BT1120_D7	CIF_D7	GPIO4_A7_d	AM27
	SPI2_CS1_M1	UART8_TX_M0	I2C6_SDA_M3		I2S1_SDI3_M0	PCIE30X2_PERSTN_M1	BT1120_CLKOUT	CIF_CLKIN	GPIO4_B0_d	AK26
SATA2_ACT_LED_M0	SPDIF1_TX_M1	SPI0_CS1_M1	UART8_RX_M0	I2C6_SCL_M3		I2S1_SDO0_M0	PCIE30X1_0_BUTTON_RSTN		MIPI_CAMERA0_CLK_M0	GPIO4_B1_u	AL24
CAN1_RX_M1	PWM14_M1	SPI0_CS0_M1	UART8_RTSN_M0	I2C7_SCL_M3		I2S1_SDO1_M0	PCIE30X1_1_BUTTON_RSTN	BT1120_D8	CIF_HREF	GPIO4_B2_u	AK25
CAN1_TX_M1	PWM15_IR_M1		UART8_CTSN_M0	I2C7_SDA_M3		I2S1_SDO2_M0	PCIE20X1_2_BUTTON_RSTN	BT1120_D9	CIF_VSYNC	GPIO4_B3_u	AM25
SPDIF0_TX_M1	PWM1_IR_M1	DP0_HPDIN_M0	UART9_TX_M1			I2S1_SDO3_M0	PCIE30X4_CLKREQN_M1	BT1120_D10	CIF_CLKOUT	GPIO4_B4_u	AL26
SATA1_ACT_LED_M0	PWM12_M1	SPI3_MISO_M1	UART9_RX_M1			HDMI_RX_CEC_M0	PCIE30X4_WAKEN_M1	BT1120_D11		GPIO4_B5_d	AJ26
SATA0_ACT_LED_M0	PWM13_M1	SPI3_MOSI_M1		I2C5_SCL_M1		HDMI_RX_HPDOUT_M0	PCIE30X4_PERSTN_M1	BT1120_D12		GPIO4_B6_d	AJ27
		SPI3_CLK_M1		I2C5_SDA_M1		HDMI_TX0_SCL_M0	PCIE20X1_2_CLKREQN_M1	BT1120_D13		GPIO4_B7_u	AJ28
		SPI3_CS0_M1		I2C8_SCL_M3		HDMI_TX0_SDA_M0	PCIE20X1_2_WAKEN_M1	BT1120_D14		GPIO4_C0_u	AJ25
SPDIF1_TX_M2	PWM6_M1	SPI3_CS1_M1		I2C8_SDA_M3		HDMI_TX0_CEC_M0	PCIE20X1_2_PERSTN_M1	BT1120_D15		GPIO4_C1_d	AK24

VCC_3V3_S0

R95032 2.2K 5% R0201 — R95033 2.2K 5% R0201

CM_RST_L
HDMI1_TX_ON_H
W-DISABLE
PCIE30x1_0_CLKREQn_M1_L
PCIE30x1_0_WAKEn_M1_L
PCIE30x1_0_PERSTn_M1_L
TYPEC0_SBU1_DC
TYPEC0_SBU2_DC
USB_HOST_PWREN_H
HDMI0_TX_ON_H
I2C7_SCL_M3
I2C7_SDA_M3
PCIE30X4_CLKREQn_M1_L
PCIE30X4_WAKEn_M1_L
PCIE30X4_PERSTn_M1_L
HDMITX0_SCL_M0
HDMITX0_SDA_M0
HDMITX0_CEC_M0

VCCIO6_1V8 — AC25 — C1909 100nF X5R 6.3V C0201 — VCC_1V8_S0

VCCIO6 — AC26 — C1908 100nF X5R 6.3V C0201 — VCC_3V3_S0

RK3588-Socket

RK3588_I(VCCIO4 Domain)

U1000I

VCCIO4 Domain

Operating Voltage=1.8V/3.3V

		SPI4_MISO_M2	PCIE30X1_1_CLKREQN_M2/	DP0_HPDIN_M2		I2C2_SDA_M4	UART6_RX_M1	GPIO1_A0_d	A24
SATA1_ACT_LED_M1		SPI4_MOSI_M2	PCIE30X1_1_WAKEN_M2	DP1_HPDIN_M2		I2C2_SCL_M4	UART6_TX_M1	GPIO1_A1_d	A25
	PWM0_M2	SPI4_CLK_M2		VOP_POST_EMPTY		I2C4_SDA_M3	UART6_RTSN_M1	GPIO1_A2_d	A26
	PWM1_M2	SPI4_CS0_M2		HDMI_TX1_SDA_M2		I2C4_SCL_M3	UART6_CTSN_M1	GPIO1_A3_d	A27
		SPI2_MISO_M0		HDMI_TX1_SCL_M2				GPIO1_A4_d	B25
		SPI2_MOSI_M0		HDMI_TX0_HPD_M0				GPIO1_A5_d	B26
		SPI2_CLK_M0		HDMI_TX1_HPD_M0				GPIO1_A6_d	C24
	PWM3_IR_M3	SPI2_CS0_M0	PCIE30X1_1_PERSTN_M2			PDM1_SDI0_M1		GPIO1_A7_u	C25
		SPI2_CS1_M0	PCIE30X4_CLKREQN_M3			PDM1_SDI1_M1		GPIO1_B0_u	C27
		SPI0_MISO_M2	PCIE30X4_WAKEN_M3			PDM1_SDI2_M1		GPIO1_B1_d	D25
		SPI0_MOSI_M2	PCIE30X4_PERSTN_M3			PDM1_SDI3_M1	UART4_RX_M2	GPIO1_B2_d	D26
SATA0_ACT_LED_M1		SPI0_CLK_M2	PCIE30X1_0_WAKEN_M2			PDM1_CLK1_M1	UART4_TX_M2	GPIO1_B3_d	D27
		SPI0_CS0_M2	PCIE30X1_0_PERSTN_M2			PDM1_CLK0_M1	UART7_RX_M2	GPIO1_B4_u	E24
		SPI0_CS1_M2	PCIE30X1_0_CLKREQN_M2/				UART7_TX_M2	GPIO1_B5_u	E25
	HDMI_RX_HPDOUT_M2/	SPDIF0_TX_M3	PCIE30X2_WAKEN_M3		MIPI_CAMERA1_CLK_M0	I2C5_SCL_M3	UART1_TX_M1	GPIO1_B6_u	E26
SATA2_ACT_LED_M1	HDMI_RX_CEC_M2	PWM3_M2	SPDIF1_TX_M0	PCIE30X2_PERSTN_M3	MIPI_CAMERA2_CLK_M0	I2C5_SDA_M3	UART1_RX_M1	GPIO1_B7_u	E27
	HDMI_RX_SCL_M2	PWM14_M2			MIPI_CAMERA3_CLK_M0	I2C8_SCL_M2	UART1_RTSN_M1	GPIO1_D6_u	F24
	HDMI_RX_SDA_M2	PWM15_IR_M3		PCIE30X2_CLKREQN_M3	MIPI_CAMERA4_CLK_M0	I2C8_SDA_M2	UART1_CTSN_M1	GPIO1_D7_u	F25

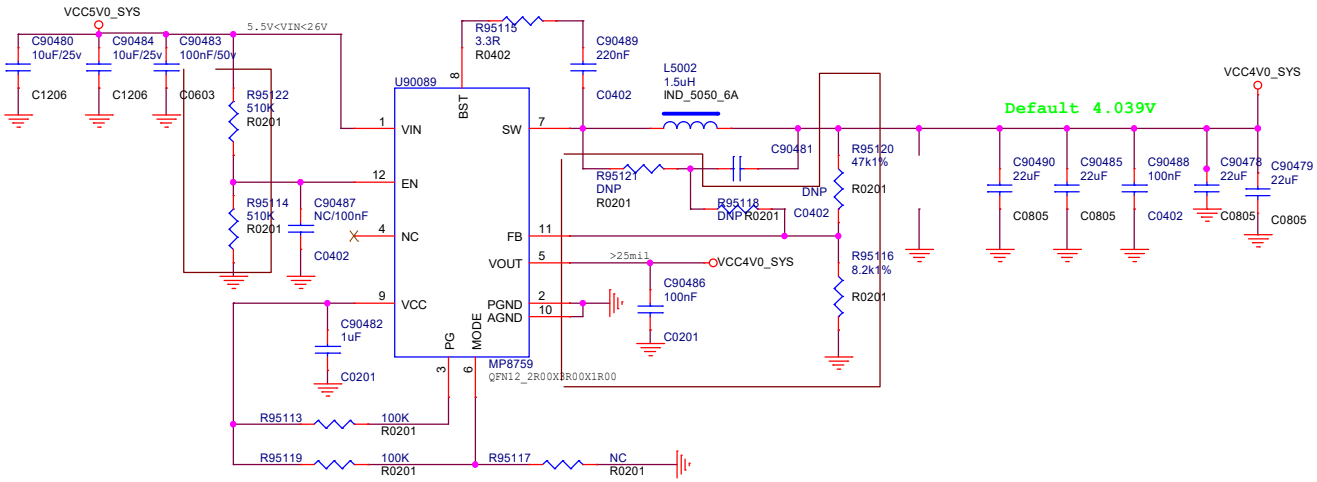
VCCIO4_1V8 — H20 — C1907 100nF X5R 6.3V C0201 — VCC_1V8_S0

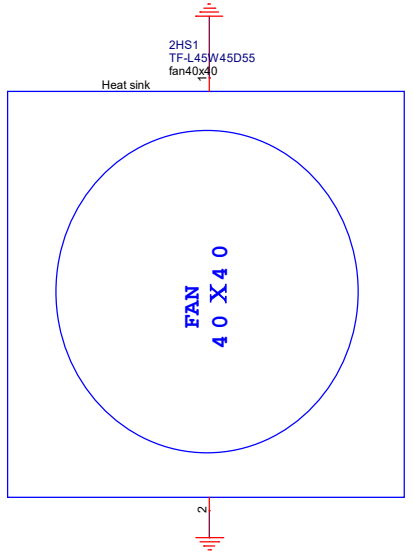
VCCIO4 — H21

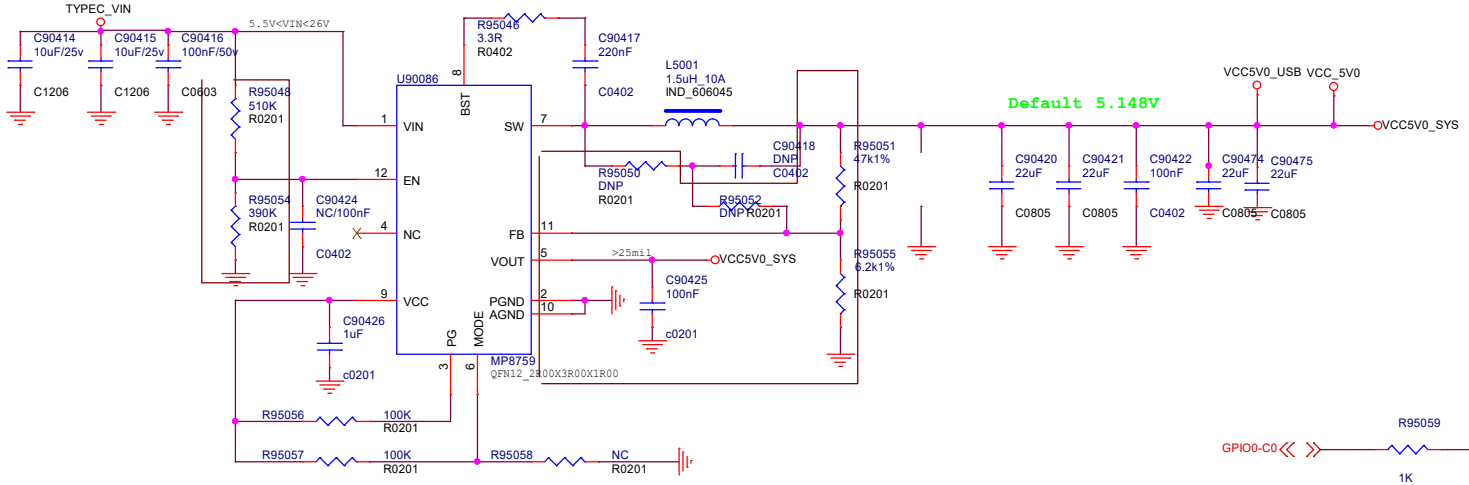
RK3588-Socket

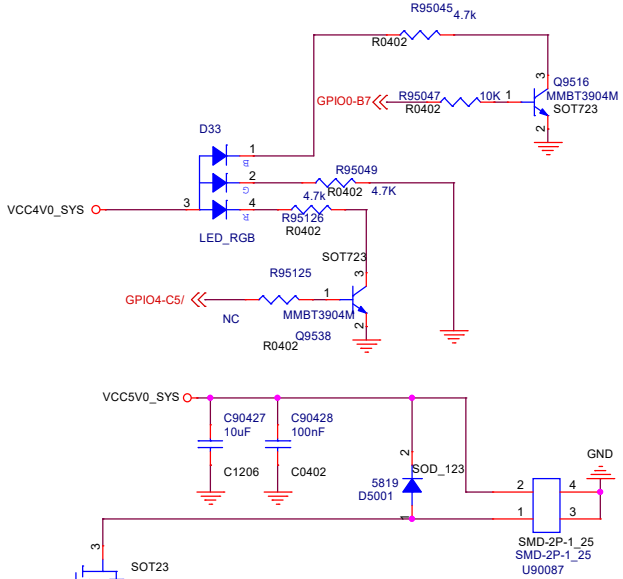
radxa			
Size: A2	Title: ROCK 5B		REV: 1.45
	Page Name: RK3588_1.8V/ 3.3V GPIO		
Date: Tuesday, April 25, 2023	Sheet 19 of 32		

TYPEC-IN ----> MP8759 DCDC 5.148V ----> USB
----> ~~VCC5V0~~ -> MP8759 DCDC 4.039V ----> PMU
----> 40PIN





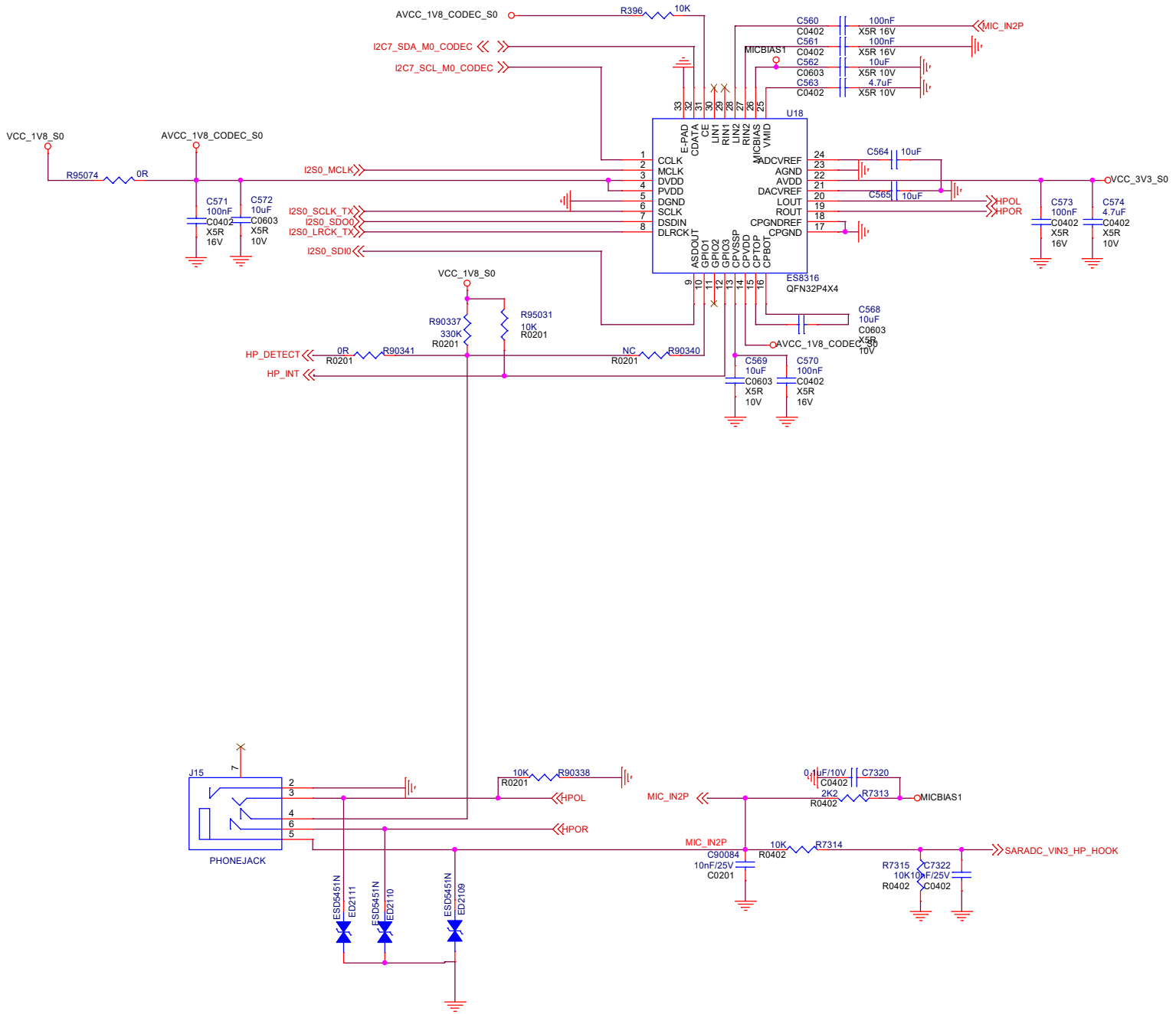




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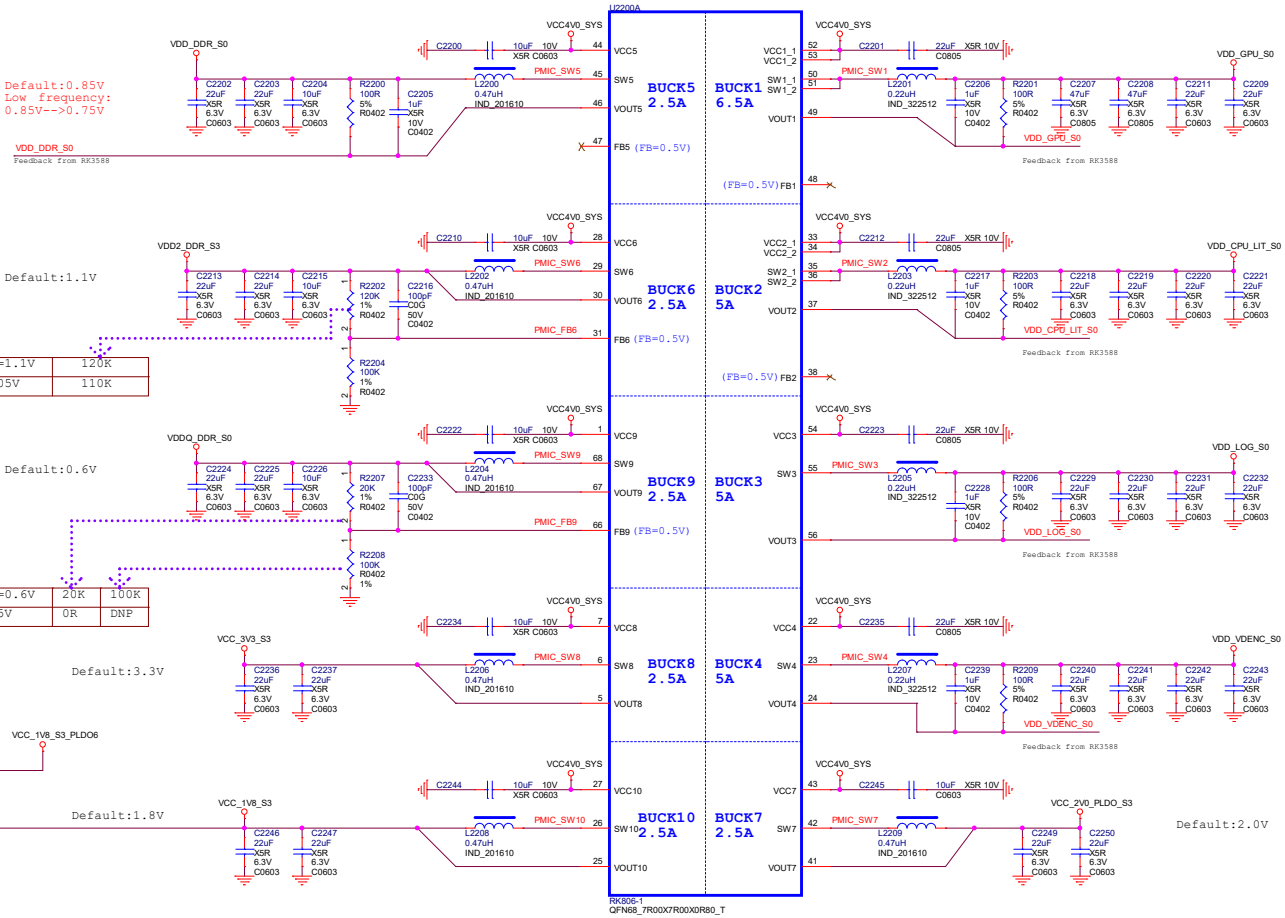
Size	Title:	REV
A3	ROCK 5B	1.45
	Page Name: RK3588_PCIE30/PCIE20/SATA30	
Date: Tuesday, April 25, 2023	Sheet 20 of 32	

CODEC ES8316



Size A3	Title: ROCK 5B	REV 1.45
	Page Name: CODEC ES8316	
Date: Tuesday, April 25, 2023	Sheet 21 of 32	

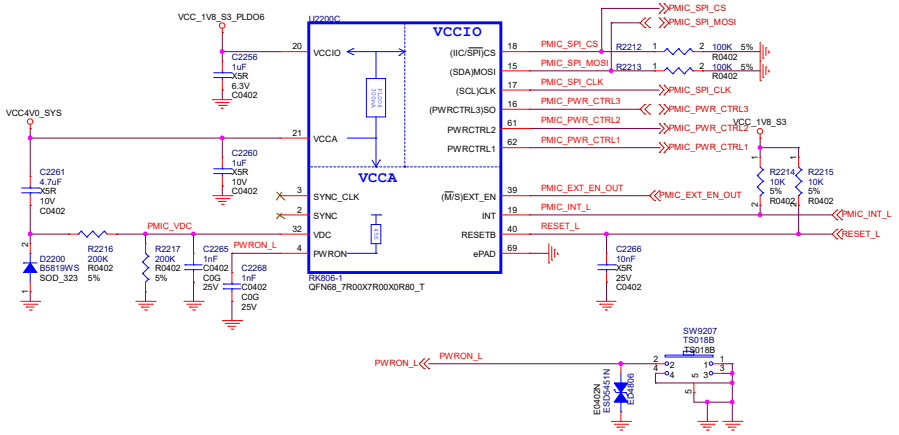
PMIC RK806-1 BUCK



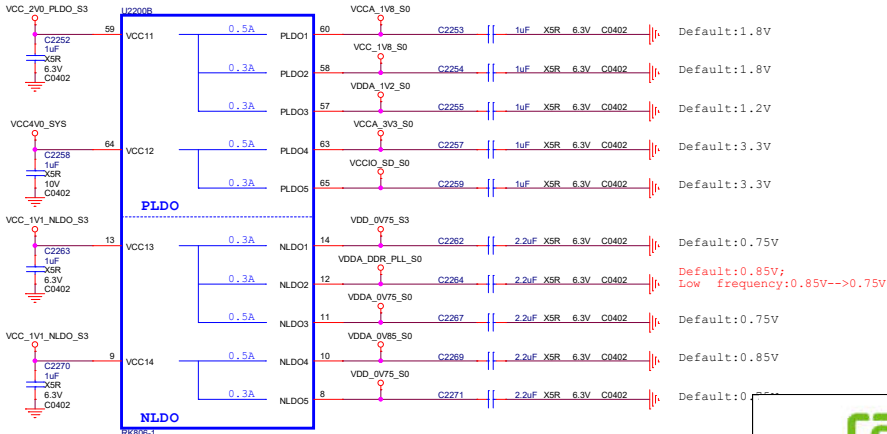




PMIC RK806-1 Managerment

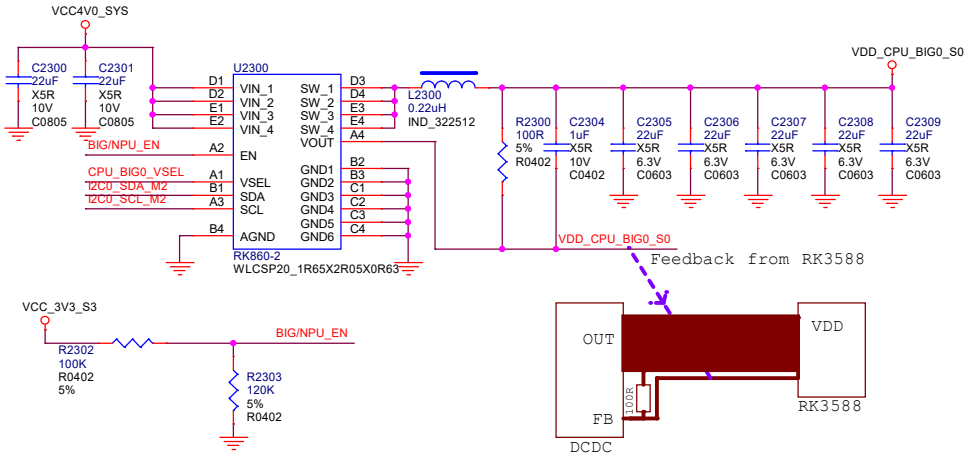


PMIC RK806-1 LDO

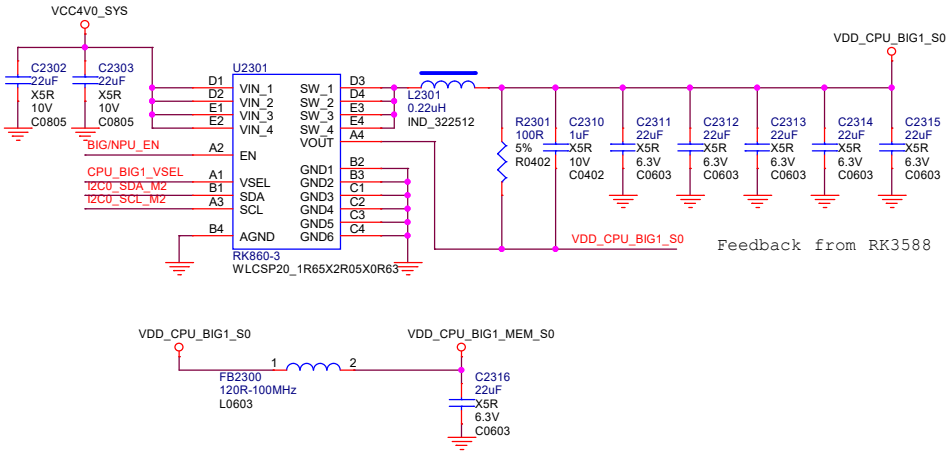


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Size	Title:	ROCK 5B		REV
A2	Page Name:	PMIC RK806-1 BUCK		1.45
Date:	Tuesday, April 25, 2023	Sheet	22 of 32	

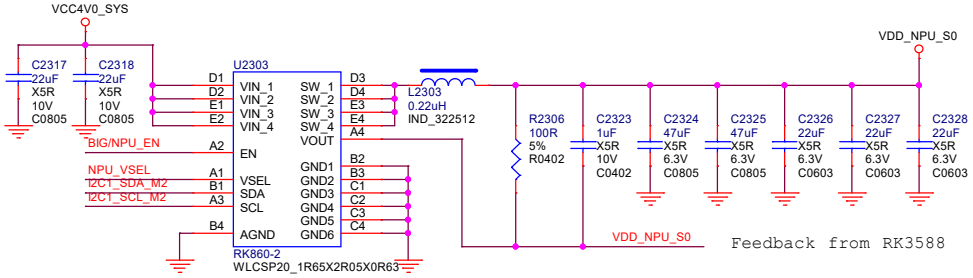
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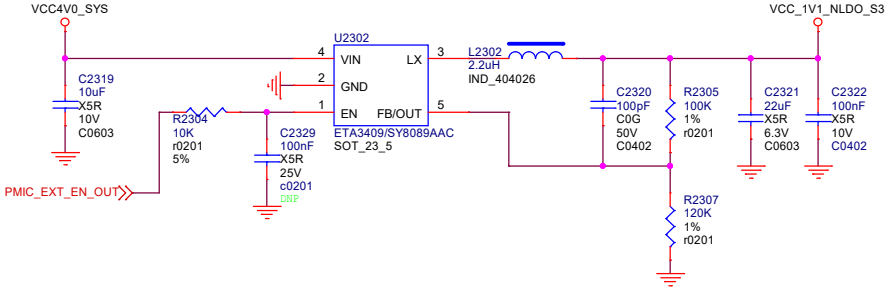
VDD_CPU_BIG1

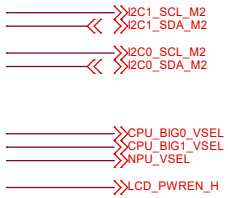


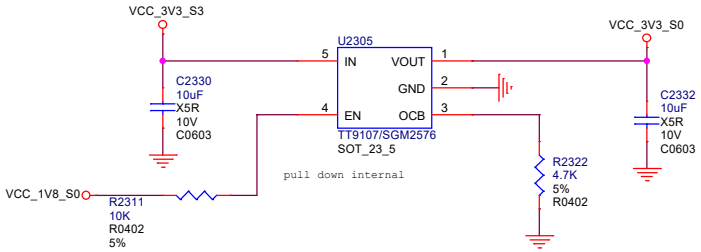
VDD_NPU

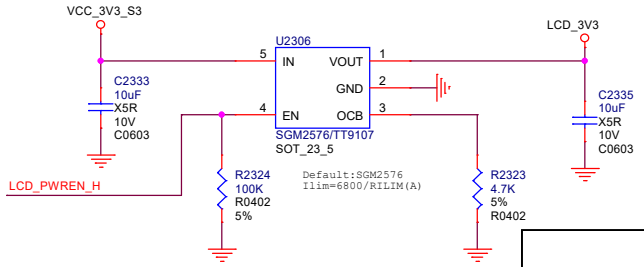


VCC_1V1_NLDO_S3



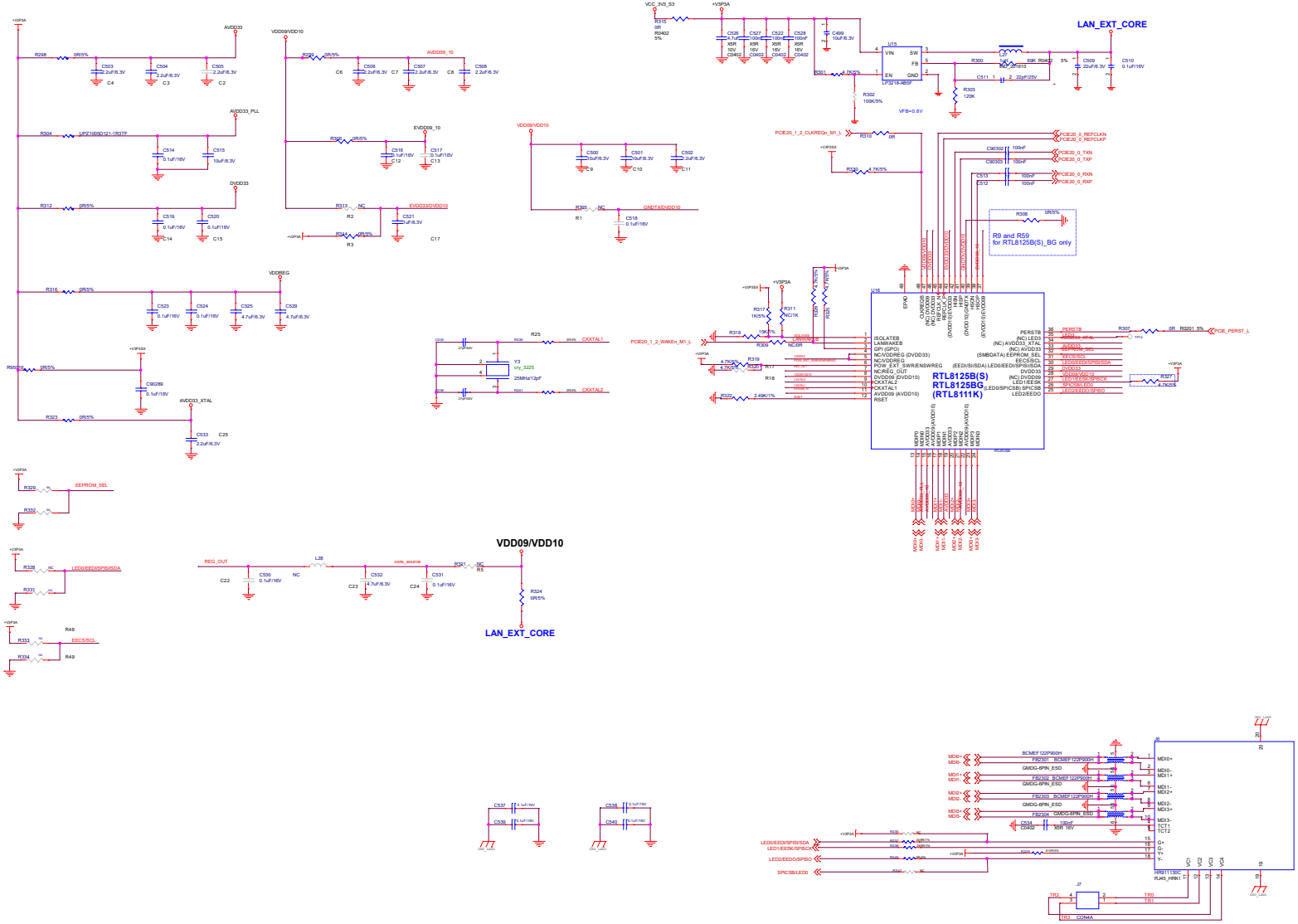






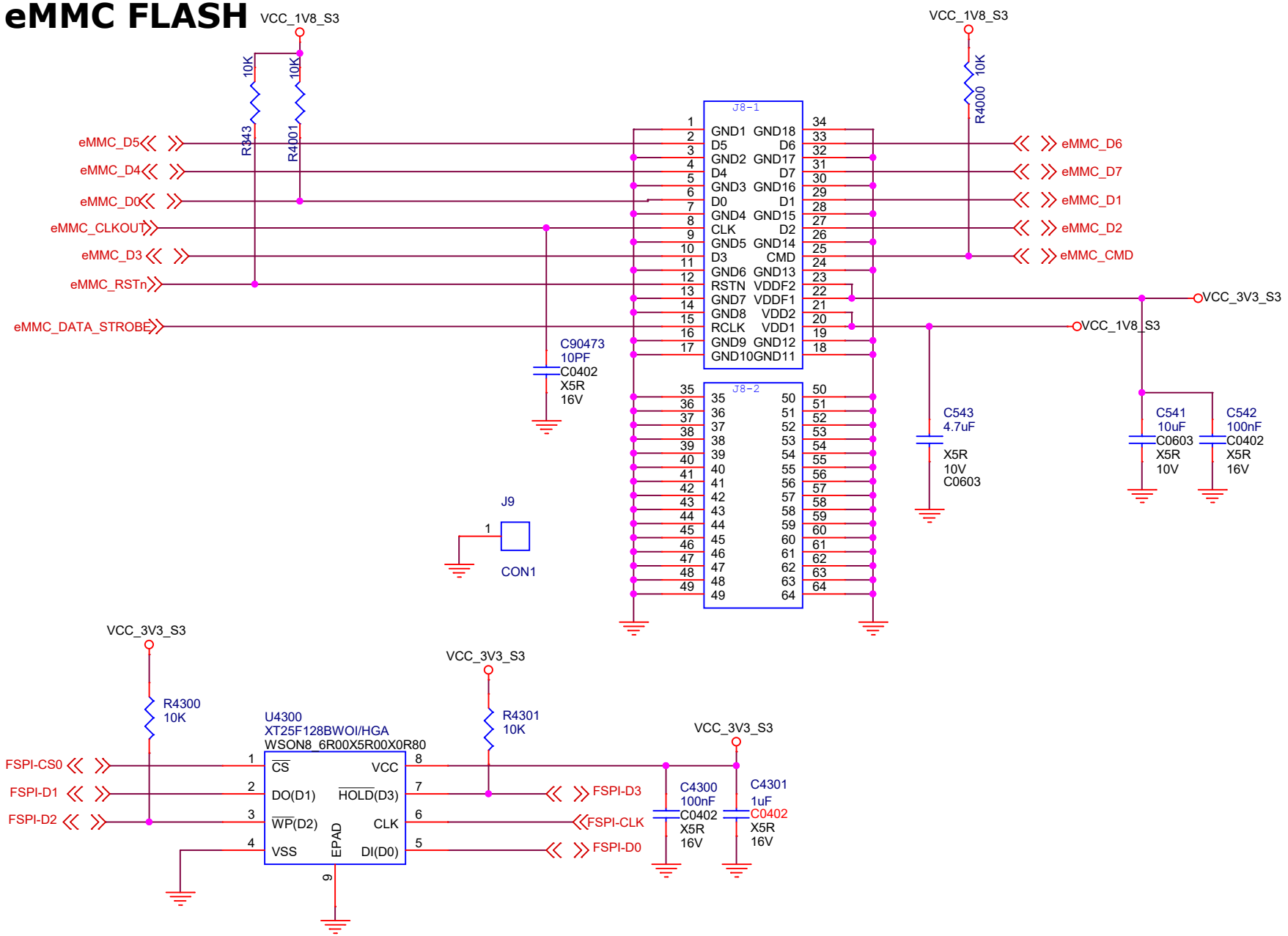
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Size	Title:	REV
A3	ROCK 5B	1.45
	Page Name: Power_Ext Discrete	
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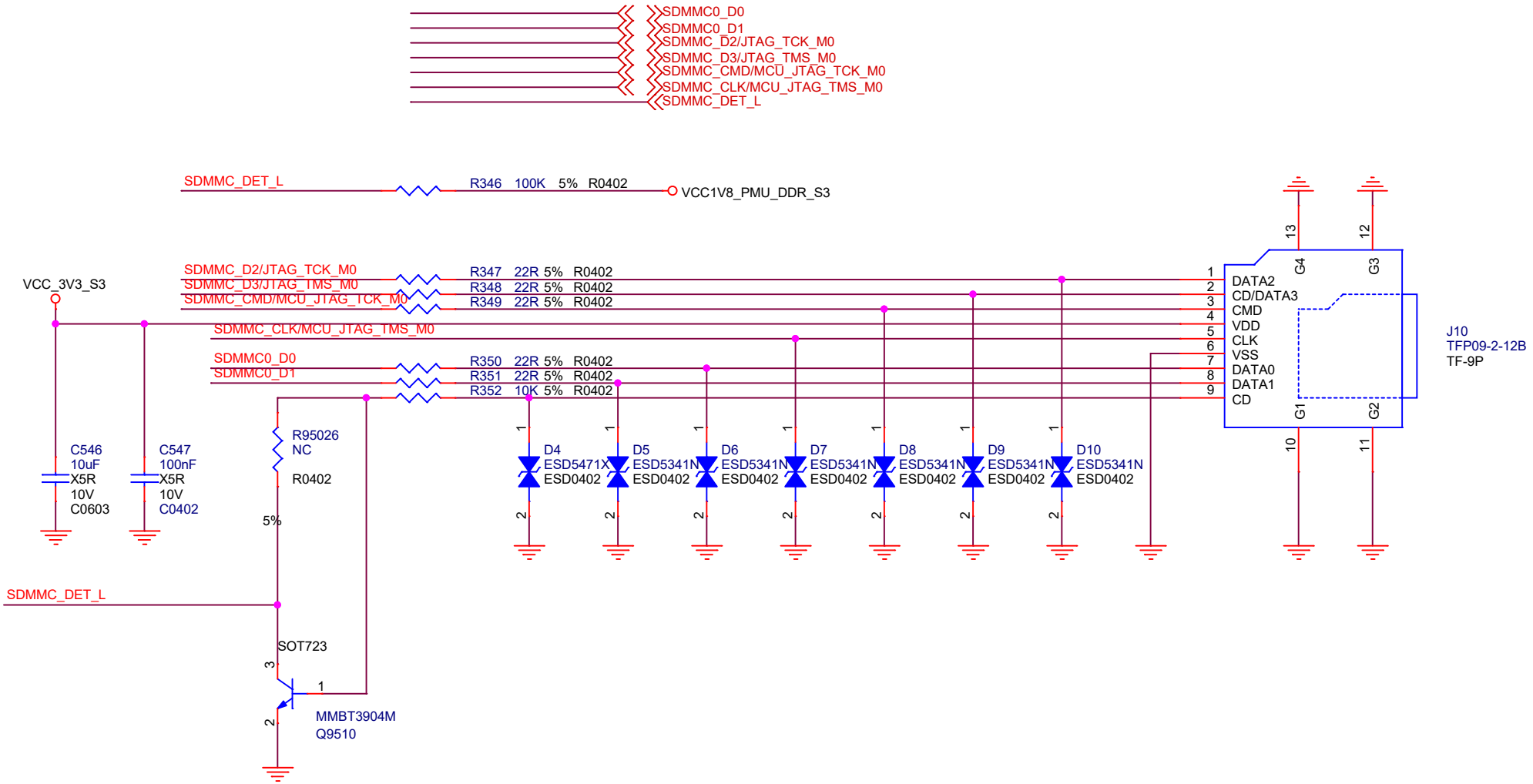
Size: A1	Title: ROCK 5B		REV: 1.45
	Page Name: PCIE TO RJ45_RTL8125B		
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eMMC FLASH



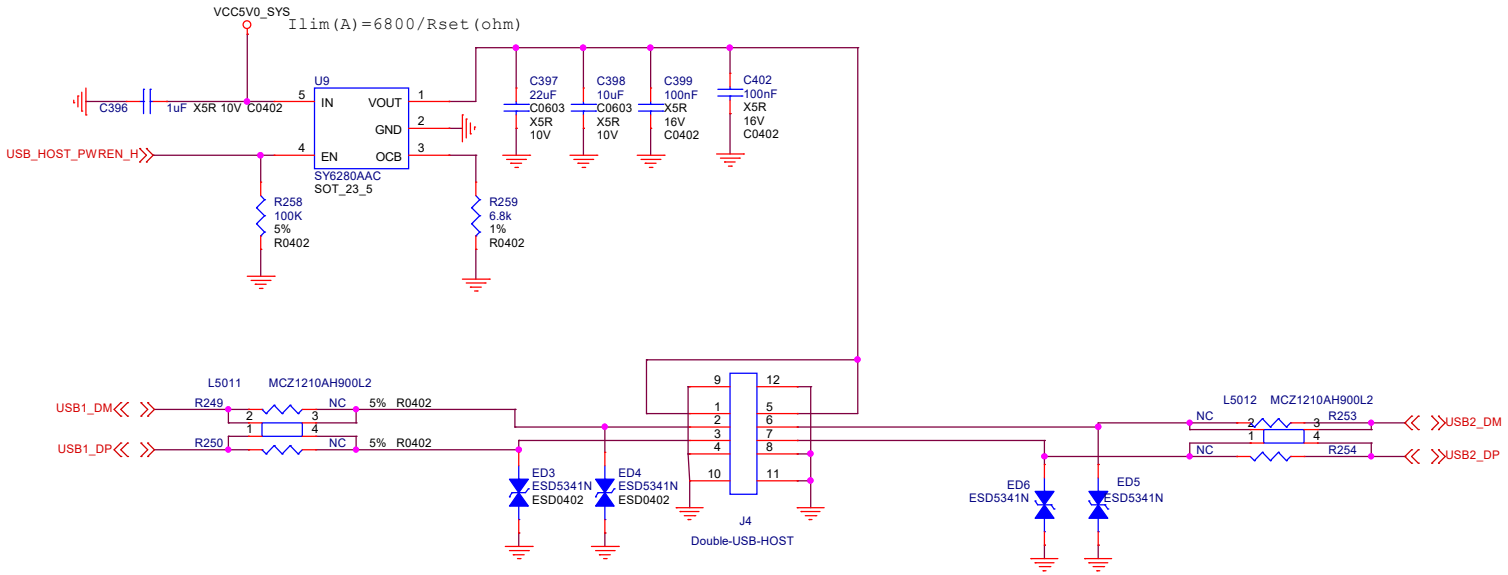
Size A4	Title: ROCK 5B	REV 1.45
	Page Name: eMMC FLASH	
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TF CARD



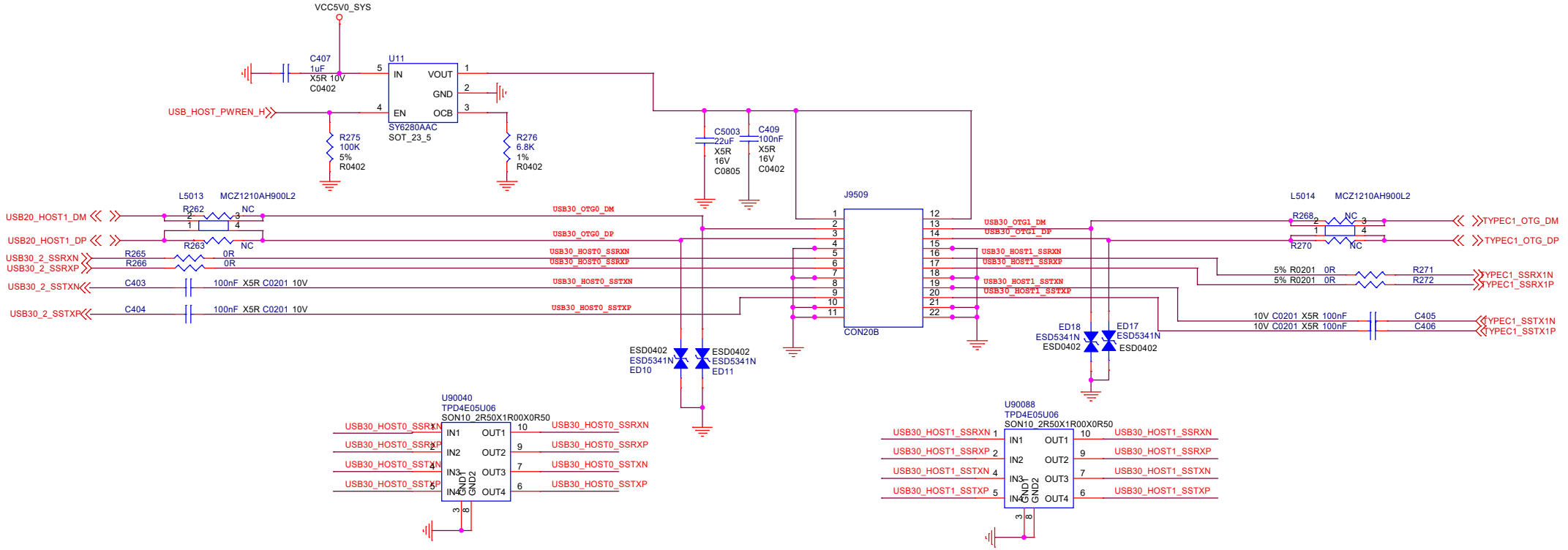
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	Page Name: TF CARD		
Date: Tuesday, April 25, 2023		Sheet 26 of 32	

USB2.0



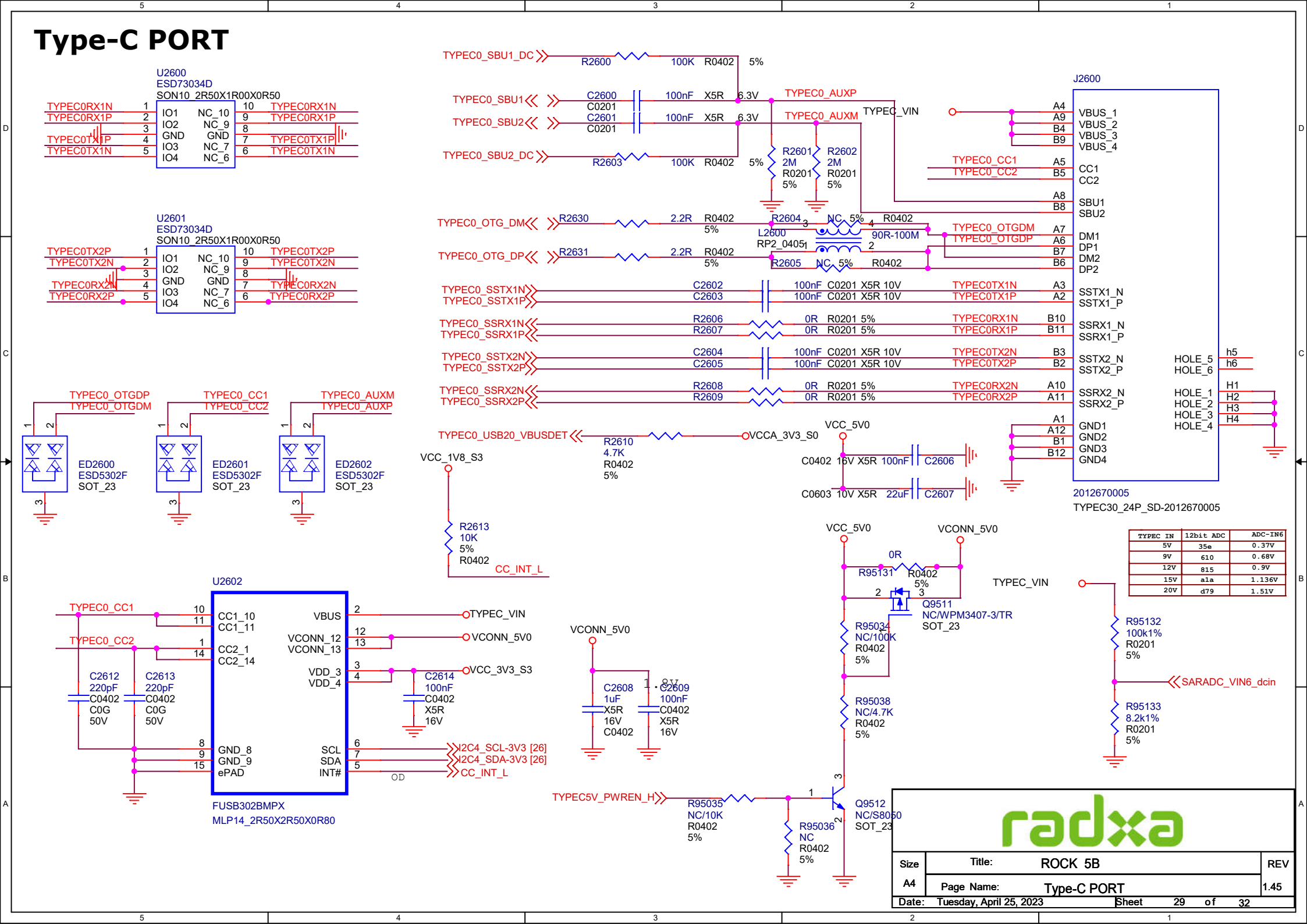
Size A3	Title: ROCK 5B	REV 1.45
	Page Name: USB2.0	
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USB3.0 HOST PORT



Size A3	Title: ROCK 5B	REV 1.45
	Page Name: USB3.0 HOST PORT	
Date: Tuesday, April 25, 2023	Sheet 28 of 32	

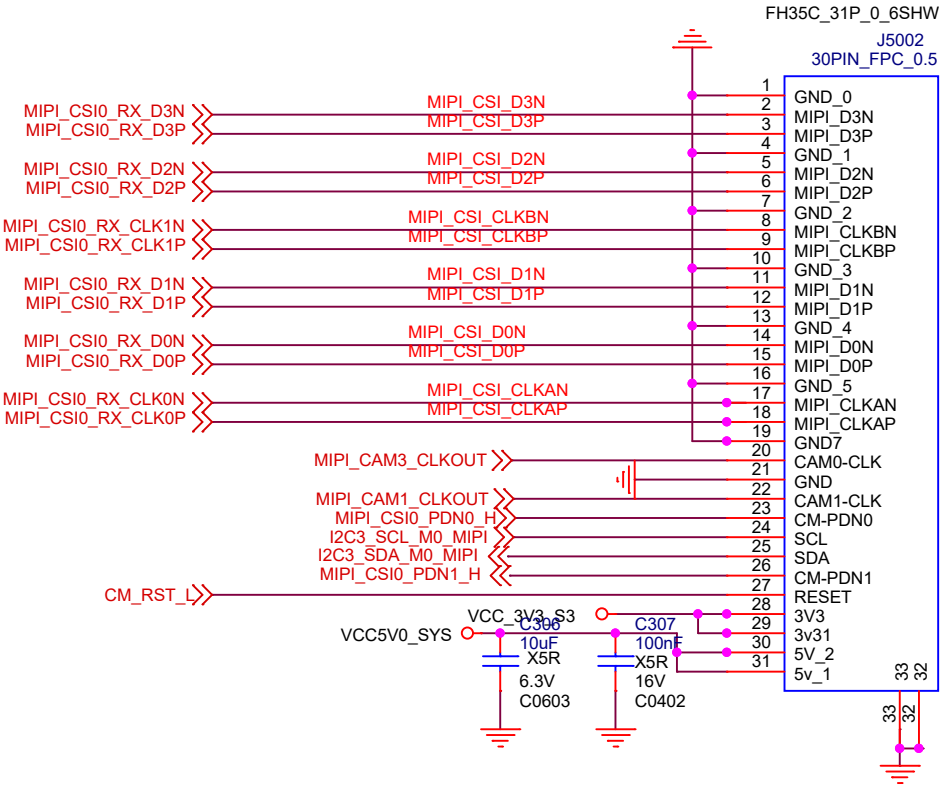
Type-C PORT

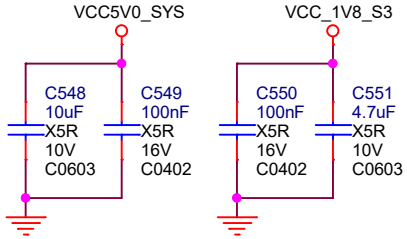


TYPEC IN	12bit ADC	ADC-IN6
5V	35e	0.37V
9V	610	0.68V
12V	815	0.9V
15V	a1a	1.136V
20V	d79	1.51V

Size A4	Title: ROCK 5B		REV 1.45
	Page Name: Type-C PORT		
Date: Tuesday, April 25, 2023		Sheet 29 of 32	

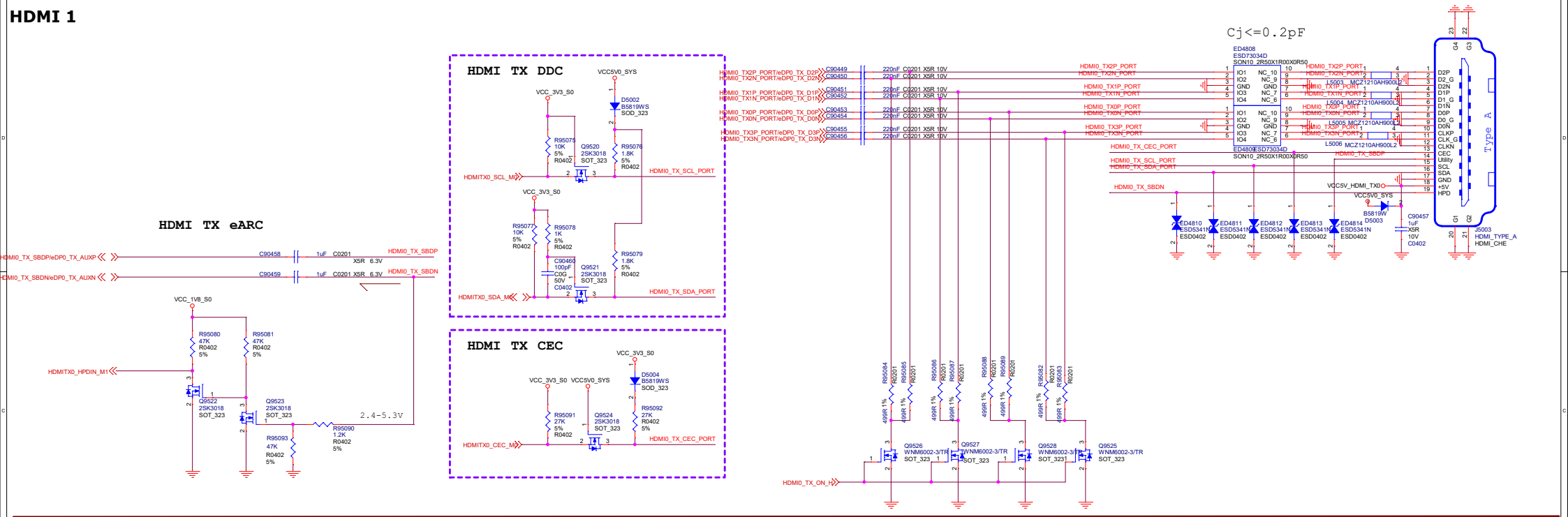
CAM



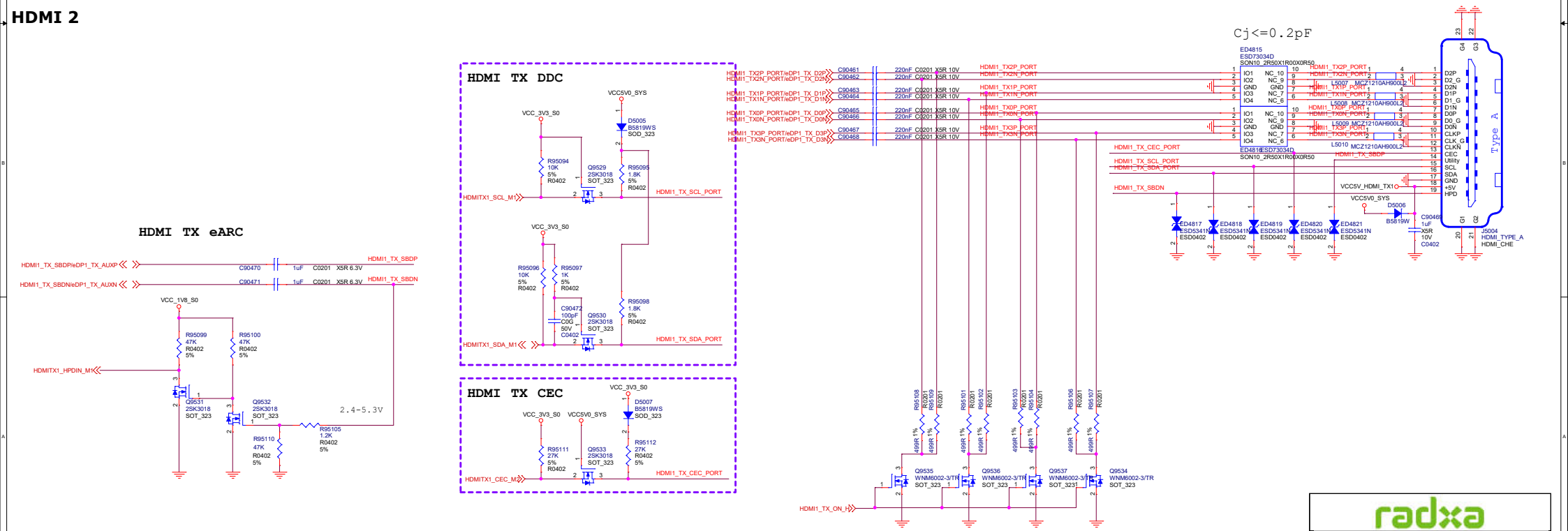


Size A4	Title: ROCK 5B		REV 1.45
	Page Name: CAM		
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HDMI 1

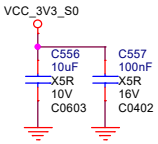


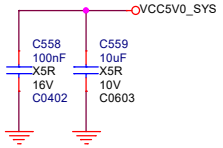
HDMI 2

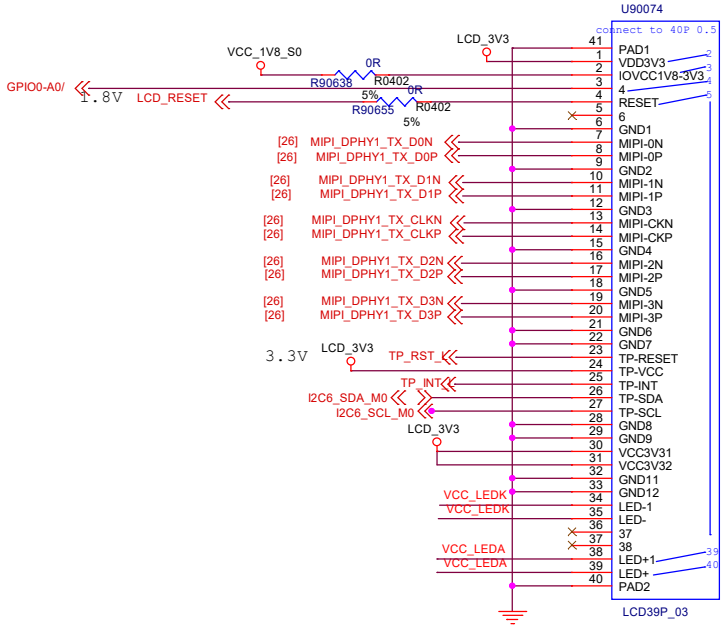


Size A2	Title: ROCK 5B		REV 1.45
	Page Name: HDMI		
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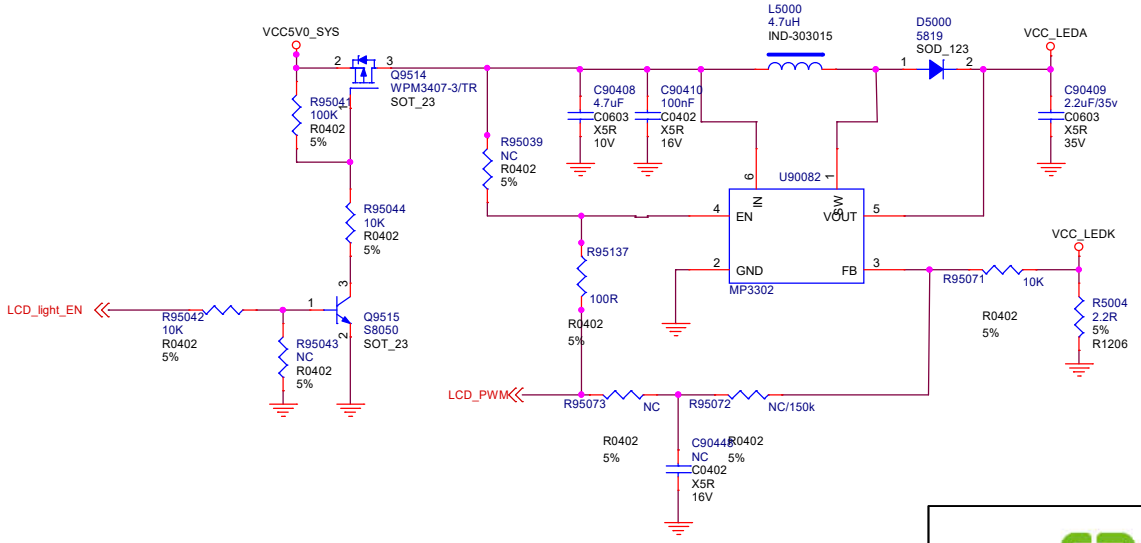
MIPI DPHY1 TX







LCD



radxa

Size	Title:	REV
A3	ROCK 5B	1.45
	Page Name: MIPI DPHY1 TX	
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